

How Costly is Empire Building? A Heterogeneous Search-and-Matching Model of Merger Markets^{*}

Fernando Anjos[†] Luiza Llussá[‡] Miguel Oliveira[§]

June 18, 2026

PRELIMINARY AND INCOMPLETE

Abstract

This paper develops a heterogeneous search-and-matching model of mergers that features two types of firms: value maximizers and empire builders. In our model, the total cost of empire building is driven by (i) the direct cost associated with the inefficient merger policies adopted by such firms; and (ii) indirect equilibrium effects. The first indirect effect mitigates empire-building inefficiencies: a merger deal involving a value maximizer and an empire builder prevents the latter from potentially engaging in deals with other empire builders, which sometimes are value-destroying. The second indirect effect amplifies empire-building inefficiencies: value maximizers rationally respond to the presence of empire builders in the market by focusing relatively more on rent-seeking (versus value creation). A calibration exercise suggests the presence of empire builders in the economy leads to a loss of 0.4-0.5% in aggregate market capitalization; with net indirect effects reducing the total cost of empire building by 14-31%.

JEL classification: D20, G32, G34, L25.

Keywords: mergers, empire-building, search-and-matching.

^{*}The authors thank conference participants at the 2026 Pitt-PSU-CMU Conference for their helpful comments. This work was funded by FCT — Fundação para a Ciência e a Tecnologia (UID/00124/2025, UID/PRR/124/2025, Nova School of Business and Economics) and LISBOA2030 (DataLab2030 - LISBOA2030 - FEDER - 01314200).

[†]NOVA School of Business and Economics. fernando.anjos@novasbe.pt

[‡]NOVA School of Business and Economics. luizallussa@gmail.com

[§]Katz Graduate School of Business — University of Pittsburgh. miguel.oliveira@pitt.edu

1 Introduction

The topic of *empire building* is pervasive in corporate finance, with several authors viewing the phenomenon as a primary source of economic inefficiency (Jensen, 1986; Moeller, Schlingemann, and Stulz, 2005). The underlying premise is that some managers either derive private benefits from running larger, more diverse firms, or are overly optimistic about M&A synergies (Malmendier and Tate, 2008).

While we acknowledge the potentially large costs associated with empire building, we argue that these costs are significantly affected by the interaction between empire builders and value-maximizing firms. In our model, these indirect effects operate through two channels. The first channel attenuates the cost of empire building: when a value maximizer successfully merges with an empire builder, the transaction removes that empire builder from the market, preventing it from engaging in future inefficient deals. The second channel amplifies the cost of empire building: empire builders are willing to accept unfavorable deal terms, which prompts value-maximizing firms to focus relatively more on rent-seeking (versus value creation).

The calibrated version of our model lends support both to the notion that the direct costs of empire building are economically relevant (about 0.4–0.5% of aggregate market capitalization) and that indirect effects matter (total costs are 14–31% smaller because of them). This latter result implies that value-maximizing behavior creates significant (net) positive spillovers on the value of inefficiently run firms, even after accounting for the negative rent-seeking channel.

We contribute to the literature not only by presenting a fresh view on the empire-building phenomenon, but also by making the broader point that in certain settings value-maximizing behavior could have positive unintended consequences that have been hitherto ignored. Since value-maximizing behavior can follow from public investments in general education and

financial literacy (Custódio, Mendes, and Metzger, 2025), it would be important to recognize that the returns to such investments are, under certain conditions, larger than they might appear (i.e., once one factors in indirect effects such as the ones we estimate).

We model an economy comprising a continuum of business units that meet randomly in a matching market and decide whether to merge. When a match occurs, two key state variables are realized, and both are observed by the matched firms. The first state variable is the level of synergies associated with combining operations; the second state variable refers to the terms of the deal, essentially the ownership share in the combined entity allocated to each of the two groups of initial shareholders. This latter aspect of our model is perhaps less common¹ and is basically a reduced-form representation of bargaining and haggling, which are important aspects of real-world merger deals. One can view our formal approach as a variation on the standard take-it-or-leave-it bargaining mechanism, with Nature presenting a deal to both sides as a take-it-or-leave-it offer. The asymmetry in surplus appropriation resulting from our mechanism can be interpreted as exogenous differences in bargaining power.

The key novel ingredient in the model is the existence of heterogeneous firm types, namely value maximizers and empire builders. While the former behave in a standard way, adopting merger policies that maximize firm value, the latter are less strict and consider accepting deals that are value-destroying, either because the deal is NPV-negative or because they are obtaining too small a share of the combined entity. The co-existence of these two types of firms generates important equilibrium effects, which matter for the ultimate total cost associated with empire building.

One indirect channel that we propose is that value maximizers provide an indirect service to the economy. Specifically, by being present in the matching market, it becomes possible that hybrid matches occur, where a deal is struck with an empire builder if synergies are high

¹See for example Smith (2006) for another exercise employing exogenously defined match payoffs.

enough. Such deals take out empire builders from the pool of merger market participants, and thus statistically prevent the occurrence of subsequent inefficient deals. In the context of our model, it is straightforward to formalize this spillover effect, which we also quantify.

Our second indirect channel is less benign. Specifically, value maximizers anticipate that matches with empire builders might occur, and in particular that such matches might be acceptable for empire builders even if the shareholders of such firms obtain a low ownership fraction of the combined entity. Importantly, this means that value maximizers have an incentive to wait for these “opportunistic” deals, which naturally comes at the expense of making deals driven solely by NPV.

Using the steady-state outcomes of the model, we conduct a simple calibration exercise. An important ingredient in our calibration is the distribution of announcement returns (for a recent critique of this approach, see Ben-David, Bhattacharya, Huang, and Jacobsen (2025)), and in particular we pick parameters to match several moments of said distribution. In order to perform the best possible match between our model and its empirical counterpart, we construct several versions of the announcement-returns distribution and discuss why some ways of cutting the data are more appropriate than others for our specific exercise.

Our paper connects with several strands of the literature on M&A and, more broadly, corporate finance. While we model empire-building behavior in reduced-form, we have in mind various mechanisms driving such behavior. For one, we build on seminal literature connecting agency theory (Jensen and Meckling, 1976) to takeover activity. Jensen (1986) is an important conceptual reference (Free Cash Flow hypothesis); with Harford (1999) and Moeller, Schlingemann, and Stulz (2005) as examples of key empirical studies consistent with an agency-theoretical approach to M&A.² While agency problems can explain inefficient mergers and in particular empire-building behavior, other mechanisms can be at play,

²More recently, Yim (2013) suggests that agency problems are worse for young executives, who have incentives to be very active in takeover markets.

namely managerial overconfidence, as shown in Malmendier and Tate (2008).³ And beyond overconfidence, unsophisticated managers might simply follow decision rules that are not appropriate, such as EPS maximization (Ben-David and Chincio, 2025), and pursue EPS-accretive deals at the expense of value creation; or, managers might just use hurdle rates (Barry, Carlin, Crane, and Graham, 2026). Our paper is related to the aforementioned work but poses novel research questions by focusing on the interaction between empire-building types and value-maximizing firms.⁴

There is also a large body of theoretical literature on M&A that our model is close to, namely search-and-matching and real-options approaches to the phenomenon: Hackbarth and Morellec (2008), Rhodes-Kropf and Robinson (2008), Albuquerque and Schroth (2015), Anjos and Fracassi (2018), David (2021), and Guadalupe, Rappoport, Salanié, and Thomas (2024) (among others). While we share the methodological style with these papers, we are the first, to the best of our knowledge, to model merger markets with the kind of heterogeneity we consider (value-maximizing and heuristic managers interacting).

Finally, there is a link between our paper and behavioral contract theory (Gervais, Heaton, and Odean, 2011; Kőszegi, 2014). In this literature, a common theme is that a rational (or sophisticated) principal is able to exploit a biased or less-sophisticated agent. While one of our indirect channels is in the spirit of this literature, we also show that in a search-and-matching setting rational types can generate positive spillovers for less-sophisticated types (even if unintentionally).

This paper is organized as follows. Section 2 describes our setup and characterizes the steady-state. Section 3 contains the data description, calibration, and quantitative implica-

³Also see Gervais, Heaton, and Odean (2011), which is not per se about M&A but does provide a model connecting managerial overconfidence and investment.

⁴There is another line of research on inefficient mergers that our work is perhaps less directly linked to, specifically papers that propose investor misvaluation as a key driver of inefficient M&A (Shleifer and Vishny, 2003; Dong, Hirshleifer, Richardson, and Teoh, 2006). In contrast, our model assumes that financial markets are efficient and correctly value firms, taking into account the type of manager who is in charge. Our model could potentially be extended to include investor misvaluation, which we leave for future work.

tions. Section 4 summarizes our results. All proofs are in the appendix.

2 Model

2.1 Setup

Our model builds on the search-and-matching framework of Anjos and Fracassi (2018), in the spirit of the seminal work in labor economics of Diamond (1993) and Mortensen and Pissarides (1994). Time is continuous. The economy is populated by a unit-mass continuum of business units (BUs). A BU can operate on its own—we call this a *small firm*—or be merged with one other unit into a two-unit entity, which we call a *big firm*. A small firm produces profit at rate π . A big firm produces profit at rate $\pi + x$ per unit, where x is a merger-specific synergy drawn from a distribution $\mathcal{X}$ with support $[\underline{x}, \bar{x}]$, $\underline{x} < 0 < \bar{x}$.⁵ Thus, both value-creating and value-destroying mergers are possible. All agents are risk neutral and discount the future at rate ρ .

Small firms meet in a merger market with constant-returns-to-scale matching technology. Matches arrive according to an exogenous Poisson process with intensity λ_0 . Big firms cannot undertake further mergers. Both small and big firms die according to an independent Poisson process with intensity λ_1 ; upon death, a BU is immediately replaced by a new unit of the same type, so the composition of the economy is stationary.⁶

A fraction θ of business units are *value maximizers* (VMs) and the remaining fraction $1 - \theta$ are *empire builders* (EBs). As is standard in these models, optimal behavior takes the form of a threshold rule: a match is converted into a merger if and only if the synergy x exceeds a cutoff x^* . Value maximizers set this cutoff optimally. Empire builders, by contrast,

⁵We abuse notation in the usual way: x denotes both the random variable and its realization.

⁶The death processes of the two BUs inside a big firm are assumed to be perfectly correlated, ensuring that both units always exit together.

are willing to accept deals that a value maximizer would reject. We capture this by assuming that the EB acceptance threshold is lower than the VM threshold by an amount $\eta \geq 0$.⁷

Mergers go through only if both sides agree to the deal. Conditional on a pair meeting, the terms of trade are determined by a random draw of the share split s^l from a distribution $\mathcal{S}$ with support $[\underline{s}, 1/2]$. After s^l is realized, each party is equally likely to receive the low share s^l or the high share $s^h \equiv 1 - s^l$. A merger occurs only if the realized terms satisfy both parties' participation constraints.⁸

Finally, the analysis that follows is restricted to the equilibrium, or steady state, of the economy.

2.2 Merger market equilibrium

2.2.1 Merger probabilities

When a match occurs, two random variables are realized: the synergy level x and the share split s^l . Denote by $x_v(s)$ the endogenous threshold above which a VM accepts a match in which it receives share s . The corresponding EB thresholds are shifted down by the constant η introduced before: $x_b(s) = x_v(s) - \eta$, where $\eta \geq 0$.

Given these acceptance thresholds, let us define the probability of a successful merger for

⁷We are deliberately agnostic about the source of this bias. It could reflect managerial overconfidence, agency problems, or a lack of financial sophistication in evaluating merger opportunities (e.g., a heuristic approach to merger decisions with a too-low hurdle rate). What matters for our analysis is that empire builders are less selective. The parameter η functions as a reduced-form device intended to encompass these different mechanisms.

⁸The exogenous share split is a reduced-form device that captures deal-specific determinants of bargaining power—urgency, advisor quality, financing constraints, legal and tax considerations, competing bidders, and negotiation skill—without the technical complications of explicitly modeling bargaining between agents who may have biased beliefs or nonstandard objectives. In particular, it avoids the need to specify the higher-order beliefs each agent holds about the other's bias or sophistication, an issue that arises, for instance, when empire building is modeled through distorted beliefs about synergies.

each type of firm (VM or EB) and for each possible share as

$$p_i(s) := 1 - F_{\mathcal{X}}(x_i(s)). \quad (1)$$

For each match, the most restrictive party sets the binding constraint. For a realized s^l , a match is then converted into a merger with varying likelihood: (i) if two EB types match (“ bb match”), then the probability is $p_b(s^l)$, since the party receiving the low share is more demanding; (ii) if two VM types match (“ vv match”), then the probability is $p_v(s^l)$; (iii) if an EB and a VM type match (“ bv match”), and the VM receives the low share, then the probability is $p_v(s^l)$, but if the VM type receives the high share, then the probability is $\min\{p_b(s^l), p_v(1 - s^l)\}$ (recall that both need to agree to the merger).⁹

2.2.2 Business unit masses

We denote by a_b and a_v the mass of single-BU firms that are EBs and VMs, respectively; and by m_b and m_v the mass of BUs operating in big firms that are EBs and VMs. Naturally, we have that

$$a_b + m_b = 1 - \theta, \quad (2)$$

and

$$a_v + m_v = \theta, \quad (3)$$

which follows from our assumption that when BUs die, they are replaced by BUs of the same type (EBs or VMs). We also denote the total mass of small firms and big firms as $a = a_b + a_v$ and $m = m_b + m_v$, respectively.

The following assumption specifies how firms are matched in the merger market.

⁹We assume that the synergy distribution is the same across the three kinds of matches.

Assumption 1 *The probability that a BU is matched with another BU of type $t \in \{EB, VM\}$ is proportional to the mass of type t in the population of small firms.*

Since the merger probability $p_i(s)$ depends on the realized share s^l (by shifting the acceptance threshold $x_i(s^l)$), the balance equations involve averaged probabilities, integrated over s . For same-type matches, these are:

$$\tilde{p}_{bb} := \int_{\underline{s}}^{1/2} p_b(s^l) d\mathcal{S}(s^l), \quad \tilde{p}_{vv} := \int_{\underline{s}}^{1/2} p_v(s^l) d\mathcal{S}(s^l). \quad (4)$$

For hybrid matches, there are two sub-cases depending on which share the VM receives. If the VM receives the low share, its threshold $x_v(s^l)$ is binding, and the average acceptance probability is simply $\tilde{p}_{vv}$. If the VM receives the high share (denoted hV), either the EB's and the VM's constraints may bind, and we define

$$\tilde{p}_{bv}^{hV} = \int_{\underline{s}}^{1/2} \min \{p_b(s^l), p_v(1 - s^l)\} d\mathcal{S}(s^l). \quad (5)$$

The overall merger probability in hybrid matches, common to both parties, is

$$\tilde{p}_{bv} := \frac{1}{2} (\tilde{p}_{vv} + \tilde{p}_{bv}^{hV}), \quad (6)$$

Let us first consider the case of EB BUs. The following balance equation needs to hold in equilibrium:

$$\underbrace{m_b \lambda_1 dt}_{\text{outflows}} = \underbrace{a_b \lambda_0 dt \frac{a_b}{a} \tilde{p}_{bb}}_{bb\text{-match inflows}} + \underbrace{a_b \lambda_0 dt \frac{a_v}{a} \left(\frac{\tilde{p}_{vv} + \tilde{p}_{bv}^{hV}}{2} \right)}_{bv\text{-match inflows}} \quad (7)$$

A parallel condition holds for VM BUs:

$$\underbrace{m_v \lambda_1 dt}_{\text{outflows}} = \underbrace{a_v \lambda_0 dt \frac{a_v}{a} \tilde{p}_{vv}}_{vv\text{-match inflows}} + \underbrace{a_v \lambda_0 dt \frac{a_b}{a} \left(\frac{\tilde{p}_{vv} + \tilde{p}_{bv}^{hV}}{2} \right)}_{bv\text{-match inflows}} \quad (8)$$

The equilibrium mass of BUs is finally obtained by solving the system of equations involving (2), (3), (7), and (8).

It is also useful to decompose m_b by match type. Let m_{bb} and m_{bv} denote the mass of merged EB BUs that are part of a same-type and a hybrid match, respectively. From the balance equations,

$$m_{bb} = \frac{\lambda_0 a_b^2}{\lambda_1 a} \tilde{p}_{bb} \quad (9)$$

$$m_{bv} = \frac{\lambda_0 a_b a_v}{\lambda_1 a} \left(\frac{\tilde{p}_{vv} + \tilde{p}_{bv}^{hV}}{2} \right) \quad (10)$$

2.2.3 Valuation and optimal thresholds

Having characterized the equilibrium masses, we turn to valuation and optimal decision-making. The value of a BU inside a big firm is denoted by $J_1(x)$. Since bargaining affects only the division of merger gains, and not the cash flows generated by the merged firm, the big-firm BU valuation does not depend on the drawn s^l and x is a sufficient statistic for firm value. In equilibrium,

$$\begin{aligned} \rho J_1(x) dt &= (\pi + x) dt + E[dJ_1(x)] \Leftrightarrow \\ \rho J_1(x) dt &= (\pi + x) dt + \underbrace{\lambda_1 dt \times (-J_1(x))}_{\text{death}} + \underbrace{(1 - \lambda_1 dt) \times 0}_{\text{continuation}} \Leftrightarrow \\ J_1(x) &= \frac{\pi + x}{\rho + \lambda_1}, \end{aligned} \quad (11)$$

and the total value of a big firm (which includes two BUs) is just $2J_1(x)$.

A value-maximizing BU that receives share s of the merged entity will accept a match if and only if its share of the big-firm value weakly exceeds its stand-alone value. At the

optimal threshold, this participation constraint holds with equality:

$$s \times 2 J_1(x_v(s)) = J_{0,v}, \quad (12)$$

which pins down the policy function $x_v(s)$. Note that since a higher share makes the merger more attractive, the required synergy is lower when the VM receives the high share: $x_v(1 - s^l) < x_v(s^l)$. As introduced before, the EB acceptance thresholds are shifted down by η :

$$x_b(s) = x_v(s) - \eta. \quad (13)$$

The stand-alone values of EB and VM types, $J_{0,b}$ and $J_{0,v}$, satisfy the following valuation equations:

$$\begin{aligned} (\rho + \lambda_1) J_{0,b} = & \underbrace{\pi + \lambda_0 \frac{a_b}{a} \int_{\underline{s}}^{1/2} \int_{x_b(s^l)}^{\bar{x}} (J_1(x) - J_{0,b}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{bb \text{ match}} \\ & + \underbrace{\lambda_0 \frac{a_v}{a} \frac{1}{2} \int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} ((1 - s^l) \times 2 J_1(x) - J_{0,b}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{bv \text{ match, VM receives low share}} \\ & + \underbrace{\lambda_0 \frac{a_v}{a} \frac{1}{2} \int_{\underline{s}}^{1/2} \int_{\max\{x_b(s^l), x_v(1-s^l)\}}^{\bar{x}} (s^l \times 2 J_1(x) - J_{0,b}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{bv \text{ match, VM receives high share}} \quad (14) \end{aligned}$$

$$\begin{aligned} (\rho + \lambda_1) J_{0,v} = & \underbrace{\pi + \lambda_0 \frac{a_v}{a} \int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} (J_1(x) - J_{0,v}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{vv \text{ match}} \\ & + \underbrace{\lambda_0 \frac{a_b}{a} \frac{1}{2} \int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} (s^l \times 2 J_1(x) - J_{0,v}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{bv \text{ match, VM receives low share}} \end{aligned}$$

$$+ \lambda_0 \underbrace{\frac{a_b}{a} \frac{1}{2} \int_{\underline{s}}^{1/2} \int_{\max\{x_b(s^l), x_v(1-s^l)\}}^{\bar{x}} ((1-s^l) \times 2 J_1(x) - J_{0,v}) d\mathcal{X}(x) d\mathcal{S}(s^l)}_{bv \text{ match, VM receives high share}} \quad (15)$$

In same-type matches (bb and vv), the symmetry of the problem faced by the two matched firms implies that the share multipliers average out across the two equally likely outcomes (receiving a low or high share), so the merger gain simplifies to $J_1(x) - J_{0,i}$. In hybrid matches, the two sub-cases must be kept separate because the binding threshold differs depending on which party receives the low share. Note also that the bv integrals carry explicit share multipliers— $s^l \times 2$ and $(1-s^l) \times 2$ —reflecting the asymmetric division of the merged firm's value.

Substituting $J_1(x)$ from (11) and collecting terms, the stand-alone values can be expressed more compactly as

$$J_{0,b} = \frac{\pi + \frac{\lambda_0}{a} \left[\frac{a_b}{\rho + \lambda_1} \mathcal{I}_{bb} + \frac{a_v}{\rho + \lambda_1} \mathcal{I}_{bv,b} \right]}{(\rho + \lambda_1) + \frac{\lambda_0}{a} [a_b \tilde{p}_{bb} + a_v \tilde{p}_{bv}]}, \quad (16)$$

$$J_{0,v} = \frac{\pi + \frac{\lambda_0}{a} \left[\frac{a_v}{\rho + \lambda_1} \mathcal{I}_{vv} + \frac{a_b}{\rho + \lambda_1} \mathcal{I}_{bv,v} \right]}{(\rho + \lambda_1) + \frac{\lambda_0}{a} [a_v \tilde{p}_{vv} + a_b \tilde{p}_{bv}]}, \quad (17)$$

where the merger-gain integrals $\mathcal{I}_{bb}$ and $\mathcal{I}_{vv}$ correspond to same-type matches:

$$\mathcal{I}_{bb} := \int_{\underline{s}}^{1/2} \int_{x_b(s^l)}^{\bar{x}} (\pi + x) d\mathcal{X}(x) d\mathcal{S}(s^l), \quad (18)$$

$$\mathcal{I}_{vv} := \int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} (\pi + x) d\mathcal{X}(x) d\mathcal{S}(s^l), \quad (19)$$

and $\mathcal{I}_{bv,b}$ and $\mathcal{I}_{bv,v}$ denote the merger gains in hybrid matches from the perspective of the

EB and the VM, respectively, averaged over the two sub-cases:

$$\begin{aligned} \mathcal{I}_{bv,b} := & \frac{1}{2} \left[\int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} (1-s^l) \times 2(\pi+x) d\mathcal{X}(x) d\mathcal{S}(s^l) \right. \\ & \left. + \int_{\underline{s}}^{1/2} \int_{\max\{x_b(s^l), x_v(1-s^l)\}}^{\bar{x}} s^l \times 2(\pi+x) d\mathcal{X}(x) d\mathcal{S}(s^l) \right], \end{aligned} \quad (20)$$

$$\begin{aligned} \mathcal{I}_{bv,v} := & \frac{1}{2} \left[\int_{\underline{s}}^{1/2} \int_{x_v(s^l)}^{\bar{x}} s^l \times 2(\pi+x) d\mathcal{X}(x) d\mathcal{S}(s^l) \right. \\ & \left. + \int_{\underline{s}}^{1/2} \int_{\max\{x_b(s^l), x_v(1-s^l)\}}^{\bar{x}} (1-s^l) \times 2(\pi+x) d\mathcal{X}(x) d\mathcal{S}(s^l) \right]. \end{aligned} \quad (21)$$

A notable feature of this system of equations is that the two types' valuations are linked through the hybrid match terms: the stand-alone value of an EB depends on the probability and terms of matching with a VM, and vice versa.

The equilibrium can be solved sequentially. For any candidate value $J_{0,v}$, the participation constraints (12) pin down the acceptance thresholds, as solving yields $x_v(s) = J_{0,v}(\rho + \lambda_1)/(2s) - \pi$, and the shift condition (13) determines the EB thresholds. The balance equations (7) and (8), and the identities (2) and (3), then determine the masses $\{a_v, a_b, m_v, m_b\}$. Substituting these thresholds and masses back into the VM valuation equation (17) closes the loop, yielding a single fixed-point condition:

$$J_{0,v} = \Phi(J_{0,v}), \quad (22)$$

where $\Phi(J_{0,v})$ denotes the right-hand side of (17) evaluated at the thresholds and masses implied by $J_{0,v}$. Once $J_{0,v}$ is determined, the EB stand-alone value $J_{0,b}$ follows directly from (16).

It is instructive to consider the limiting case of equal bargaining power, $\underline{s} \rightarrow \frac{1}{2}^-$. Proposition (1) summarizes the main results.

Proposition 1 *When $\underline{s} \rightarrow \frac{1}{2}^-$, the exercise threshold for VM firms is independent of θ and η . As such, if firms always receive an equal split, the valuation of small firms of VM type is also independent of θ and η .*

An interesting feature of the model is that under equal splits, VM optimal behavior is independent of EB presence. When each party receives exactly half the merged firm's value, only the realized synergy x matters for the VM's acceptance decision, and the binding constraint is the same regardless of the partner's type. With unequal splits, by contrast, the share the VM receives depends on which side of the bargain it occupies, and whether the partner accepts depends on the partner's threshold. When bargaining is introduced, VMs therefore have an incentive to: (i) accept value-destroying deals when compensated by a sufficiently large share, and (ii) wait for an EB partner willing to accept a smaller share (effectively being exploited by the VM). While the full characterization of these effects is presented in our calibration exercise, this result implies that bargaining power asymmetry is a necessary condition for VMs to exploit EBs' inefficient merger policies.

2.3 Comparative statics

In this section, we provide some comparative statics. For this exercise, we assume that synergies are normally distributed, $\mathcal{X} \sim \mathcal{N}(\mu_x, \sigma_x^2)$, truncated to the support $[\mu_x - 4\sigma_x, \mu_x + 4\sigma_x]$, with $\mu_x = 0$ and $\sigma_x = 0.10$. The share distribution $\mathcal{S}$ follows a triangular distribution on $[\underline{s}, 0.5]$ with mode 0.5.

[Insert Figure 1 here]

We start by showing how the merger probabilities and firm masses change for varying degrees of VM pervasiveness. Figure 1 plots the results. Panels A to C show the equilibrium

probabilities and masses assuming an equal split, i.e., $\underline{s} \rightarrow 0.5^-$, while Panels D to F repeat this exercise for random splits, with $\underline{s} = \frac{1}{3}$. By Proposition 1, the acceptance thresholds of VMs are invariant to θ , so the probability of a match originating a successful merger is invariant to the fraction of VMs in the economy, as shown in Panel A. However, since VMs are more “choosy” than EBs, their presence reduces the overall probability of a merger, and shapes the composition of firms in the economy. As we move away from a pure-EB economy and empire builders are replaced by value maximizers, EBs find it increasingly more difficult to complete low-synergy mergers, since their partners are more likely to be VMs who reject such deals. In fact, for low enough levels of θ , the mass of small firms of EB type increases, even though the total mass of EB firms drops. Moreover, the more stringent acceptance rule of VMs has a clear compositional effect on the pool of available candidates, such that for $\theta \approx 0.35$, half of the small firms are VMs, as shown in Panel B. Panel C also shows that when VMs are introduced in the economy, EBs are disproportionately absorbed by VMs in efficient hybrid matches.

Panel D in Figure 1 shows that the introduction of random splits lowers the probability of a merger. However, in contrast to Panel A, the probability of a merger is increasing in θ in the presence of bargaining power asymmetry, i.e., firms lower their acceptance thresholds when value maximizers become more common. In other words, as the ability of extracting rents from EBs decreases, being selective becomes costlier, and the probability of accepting a deal rises. Interestingly, note that the effect is asymmetric, leading to a compositional effect on the types of merged firms. As $\theta \rightarrow 1$, the probability of a hybrid match approaches the result found in an economy with equal splits. However, this result does not hold for the probability of a pure-VM or pure-EB merger, which are significantly below those found in an equal-split economy. Thus, hybrid matches become relatively more frequent (albeit the mass of EBs is low), as VMs benefit disproportionately from favorable share draws in hybrid matches (Panel F).

[Insert Figure 2 here]

Figure 2 presents a more detailed characterization of how VM prevalence (θ), the degree of EB inefficiency (η), and bargaining power asymmetry ($\underline{s}$) jointly shape VM acceptance thresholds. Panels A and B plot the acceptance threshold for varying levels of θ , while Panels C and D plot the acceptance threshold for different levels of η , with both thresholds evaluated at $s^l = \underline{s}$. As the potential merger splits become more unequal ($\underline{s}$ decreases), VMs respond by decreasing their acceptance threshold when receiving a high share, while increasing the threshold when receiving a low share (naturally, these two thresholds converge to the same value as $\underline{s} \rightarrow 0.5^-$). VMs are then more willing to accept bad deals, as long as they capture most of the merged firm value, but less willing to do so if they are expected to receive a low share. In Panels A and B we show that the optimal thresholds are shifted downward whenever θ is high: since the likelihood of being matched with an exploitable EB is low, the expected value of future deals goes down, and VMs optimally relax their acceptance thresholds. A similar effect occurs in Panels C and D in relation to the magnitude of η . As η increases, the scope of exploitation increases, and the acceptance thresholds shift upward, i.e., VMs become more selective.

[Insert Figure 3 here]

The previous figure shows the response of VMs evaluated at the most unequal share, $s^l = \underline{s}$. It is also instructive to consider what happens to the optimal threshold evaluated at an equal split, but in an economy with high bargaining power asymmetry (low $\underline{s}$). Figure 3 presents the results for $x_v (s^l \approx 0.5)$.¹⁰ Panel A shows how this threshold changes for varying levels of $\underline{s}$ and different levels of VM pervasiveness (θ), while Panel B repeats this exercise

¹⁰Note that in this case, the high share and low share threshold coincide.

for different levels of η . Naturally, as the bargaining power asymmetry decreases (higher $\underline{s}$), the acceptance threshold converges to the same value, by Proposition 1. Furthermore, Panel A shows that when VMs are rare, their acceptance threshold decreases with $\underline{s}$: if potential splits are more unequal, the scope for exploitation is greater, which increases the value of waiting. However, as merger market participants become more selective (more VMs), pure bargaining frictions reduce the likelihood of a successful merger, and VMs optimally decrease the acceptance threshold for low $\underline{s}$. Panel B shows a similar effect in relation to η : in a merger market with highly inefficient EBs, i.e., high η , the scope for exploitation is high, so the optimal acceptance threshold increases with bargaining power asymmetry. However, when EBs are similar to VMs in their policy functions, higher bargaining power asymmetry has a similar effect to lower λ_0 , and the optimal acceptance threshold falls.

[Insert Figure 4 here]

Figure 4 conducts another comparative statics exercise using the distributional features assumed in this section. The top panels plot the variation in the acceptance thresholds (high and low share) when we vary the center of the synergy distribution $\mathcal{X}$. Intuitively, firms become more selective as the average synergy improves. The top panels contain the extreme case where even the highest realization of x is (weakly) negative. In this case, there are no mergers (EBs are of a moderate type, meaning that η is not high enough to generate mergers in equilibrium). The bottom panels of the figure have the usual real-options intuition: the higher the volatility of the underlying asset (here, synergies x), the higher the (merger) option value embedded in firm valuation and the more selective VMs become.

We next show how the value of small firms and VM policy functions depend on the fraction of VM types in the economy.

[Insert Figure 5 here]

Figure 5 plots the value of small BUs, J_0 , as a function of θ . In Panel A, we shut down the bargaining channel, i.e., we solve the model with $\underline{s} \rightarrow 0.5^-$. The value of VM types themselves does not change, following the results in Proposition 1. In contrast, the value of EB types increases in θ and in particular converges to the value of VM types as θ approaches 1. As θ increases, it becomes less and less likely that an EB type will be paired with another EB type, and so their merger policies have little bite in equilibrium (since VM types are more strict). In short, as we replace EB types with VM types (e.g., by increasing financial sophistication and/or reducing agency problems), the economy benefits not only directly, but also indirectly via the improvement in the value of the remaining EB types. The figure also illustrates a partial way to quantify value-maximization spillovers in the model. For example, if $\theta = 0.5$, then this level of presence of VM types makes the valuation of (small) EB types be approximately 2% ($12.9237/12.6616 - 1$) higher than if they were operating in an economy with EB types only. Naturally, what is missing is to integrate into the spillovers measure the aggregate mass of EBs (for very high θ , the impact per EB firm is very high, but there are very few EB firms), which we do later.

Panel B shows that in the presence of bargaining power asymmetry both VM and EBs types valuation can fall. In this case, we set $\underline{s} = 1/3$. First, notice that for this calibration, where $\underline{s} = 1/3$, the value of VM firms monotonically decreases in θ ; but if VM types are not pervasive enough, i.e., θ is low enough, the valuation of VM types can be higher than in Panel A. When $\underline{s} < 1/2$, the share split is no longer fixed: in a bv match, the VM may receive the high share $(1 - s^l)$ while the EB receives the low share s^l . Because the EB's acceptance threshold is shifted down by η , the EB agrees to deals that the VM would reject if roles were reversed. This asymmetry allows VMs to capture on average a disproportionate share of the merger surplus at the expense of EBs, an effect that is entirely absent when

$\underline{s} \rightarrow 0.5^-$. As θ increases, EBs become scarce and VMs are increasingly matched with other VMs, where no such exploitation is possible, and $J_{0,v}$ falls.

Thus, in contrast to Panel A, where VM types can only positively affect EBs by preventing bad deals from happening, Panel B shows that in a more general setting, exploitation by rational agents can fully offset those gains to EBs. As θ rises, a given EB is more likely to be paired with—and exploited by—a VM. This exploitation effect is further amplified by the fact that VMs endogenously become more selective, waiting for realizations of s^l where they capture a larger share of the surplus.

2.4 Announcement returns

Next we turn to the equilibrium distribution of announcement returns implied by the model, which is formalized in Lemma 1 and illustrated in Figure 6. The features of the empirical distribution are used later in the calibration exercise.

Lemma 1 *The distribution $f(\cdot)$ of combined merger announcement returns r_{ik} , involving BU pair (i, k) with type $(i, k) \in \{EB, VM\}^2$, and formally defined as*

$$r_{ik}(x) = \frac{2J_1(x) - J_{0,i} - J_{0,k}}{J_{0,i} + J_{0,k}}, \quad (23)$$

is a mixture of three component distributions:

$$f(r) = q_{vv} f_{vv}(r) + q_{bb} f_{bb}(r) + q_{bv} f_{bv}(r), \quad (24)$$

with weights

$$q_{vv} = \frac{m_{vv}}{m}, \quad q_{bb} = \frac{m_{bb}}{m}, \quad q_{bv} = \frac{2m_{bv}}{m}, \quad (25)$$

The supports of the three component distributions are:

$$f_{vv} : \text{support} \left[0, \frac{J_1(\bar{x}) - J_{0,v}}{J_{0,v}} \right] \quad (26)$$

$$f_{bb} : \text{support} \left[\frac{J_1(x_v(0.5) - \eta) - J_{0,b}}{J_{0,b}}, \frac{J_1(\bar{x}) - J_{0,b}}{J_{0,b}} \right] \quad (27)$$

$$f_{bv} : \text{support} \left[\frac{2 J_1(x_{bv}^{\min}) - J_{0,b} - J_{0,v}}{J_{0,b} + J_{0,v}}, \frac{2 J_1(\bar{x}) - J_{0,b} - J_{0,v}}{J_{0,b} + J_{0,v}} \right], \quad (28)$$

where

$$x_{bv}^{\min} := \min_{s^l \in [\underline{s}, 0.5]} \max\{x_b(s^l), x_v(1 - s^l)\}. \quad (29)$$

[Insert Figure 6 here]

There are three types of merger that can occur, depending on the type of match: either both types inside the merged entity are VM, or both types are EB, or the combined firm is hybrid. Conditioning on x , the value of the combined firm is independent of the BU types that constitute the firm; it is simply $2J_1(x)$. However, the announcement returns will be different, since the initial stand-alone value of the small firms that are merged is different: $2J_{0,v}$ if both BUs are VMs, $2J_{0,b}$ if both BUs are EBs, and $J_{0,v} + J_{0,b}$ in the case of hybrid mergers. The overall distribution of announcement returns contains these three types of deals.

Panel A of Figure 6 illustrates the equal-split case ($\underline{s} \rightarrow 0.5^-$). The distribution is the stacking of three components with distinct supports. The left tail is generated entirely by bb mergers: since EBs accept deals with negative synergies, these mergers extend into negative return territory. Nonetheless, this same type of merger can generate high announcement returns: when two EB types merge but the realized x turns out to be large, the market is

more surprised than if, say, the two firms were VMs. Still focusing on Panel A, deals where both firms are of VM type correspond to lower surprise and so the support extends less to the right than deals where one VM and one EB are involved. Moreover, the support for this vv matches is bounded below by zero, as this type of match must at least generate a combined firm with the same valuation as the sum of small-firm valuations. While mergers of two VMs always have (weakly) positive announcement returns, hybrid matches with equal splits result in strictly positive announcement returns: VMs can never negatively surprise the market, but EBs always positively surprise the market in this type of merger (since it is compared with the low expected deal quality for an EB).

Turning to Panel B ($\underline{s} = 1/3$), introducing bargaining power asymmetry changes the return distribution in three ways. First, VMs become more selective, anticipating the option to receive a larger share in matches with EBs, so accepted vv deals place more mass on higher announcement returns (vv mergers must achieve even higher synergy levels now). Second, hybrid matches can now generate negative announcement returns: some matches will destroy value, with VMs exploiting the less restrictive merger policies of EBs. Third, due to the exploitative nature of hybrid matches, the average bb match now positively surprises the market.

3 Quantitative Exercise

3.1 Data

We employ data from the Thomson Reuters Securities Data Company (SDC) database to retrieve information from mergers and acquisitions (M&A) deals, matched with stock data from the Center for Research in Security Prices (CRSP) and financial data from Compustat. Our sample includes deals with announcement dates between January 1, 1980 and December

31, 2025. We follow Ben-David, Bhattacharya, Huang, and Jacobsen (2025) and further apply the following restrictions: (i) the bidder is a U.S. company, (ii) the bidder and the target are publicly traded firms, (iii) the deal is not classified as a spin-off, repurchase, self-tender, recapitalization, acquisition of partial or remaining interest, and the bidder is not a leverage buyout firm, (iv) the percentage of shares acquired (or sought, for non-completed deals) is at least 50%, (v) the percentage of shares held by the bidder six months before the announcement is less than 50%; (vi) drop duplicate deals by the same bidder and target on the same day; (vii) both the bidder and the target’s stock price data are available from CRSP for at least 100 trading days prior to the announcement, and accounting data for the year-end immediately prior to the announcement are available from Compustat; and (viii) the deal value is non-missing in the SDC database.

After applying these restrictions, we end up with a sample of 5,850 deals (including both completed and non-completed). Importantly, we are also interested in the target’s cumulative abnormal returns as we compute the combined cumulative abnormal returns (CARs) of the deal. Hence, our final sample considers deals that have identifiers for both the target and the acquirer, for a total of 7,217 unique firms. Appendix B provides a detailed summary of the sample construction.

We compute the CARs for each company around the announcement date, as well as the combined CARs, value-weighted by their market capitalization one month prior to the deal announcement date. CARs are calculated as returns in excess of those predicted by a market model, whose parameters, α and β , are estimated using the CRSP value-weighted index over a period from 361 until 61 trading days before the announcement. We use an event window of $[-1,1]$ around the announcement day as our preferred specification. Nonetheless, following Ben-David, Bhattacharya, Huang, and Jacobsen (2025), we perform robustness checks employing different definitions of the event window, including $[-5,5]$ around the announcement day, and a window of $[\text{Announcement}-2, \text{Close}+2]$.

3.2 Summary statistics

Table 1 presents the averages of CARs for the acquirer, the target, and the combined firms across different event windows. Panel A shows that the average (median) abnormal return over the $[-1,1]$ event window is -1.3% (-0.9%) for acquirers and 21.1% (16.6%) for targets. We compute the combined CAR as a weighted average of the individual CARs using the corresponding market capitalizations one month prior to the announcement. The combined abnormal return is, on average, 1.8%, with a median of 1%.¹¹ It is particularly noteworthy that about 40% of deals in our sample have a negative combined CAR. While a majority of deals reduce acquirer shareholder wealth (about 60%), only about 13% of deals generate negative abnormal returns for target shareholders over the shorter $[-1,1]$ window.

[Insert Table 1 here]

Panels B and C show that, as the event window is extended, acquirer average abnormal returns decrease while target abnormal returns increase. Over the $[-5,5]$ event window, acquirer abnormal returns average -1.5%, while target average CARs are about 23.5%. For the combined firms, the average CAR remains 1.8%. Finally, over the longest horizon, acquirers exhibit an average CAR of -4.4%, while targets continue to exhibit a positive and large average CAR of about 25.4%. Importantly, over this longer horizon, the combined firm exhibits a negative average CAR (-0.6%), and a majority of deals (about 51%) have a negative CAR.

[Insert Figure 7 here]

¹¹These figures are consistent with those reported by Dessaint, Eckbo, and Golubov (2025), who find, over the period 1981-2017, an average combined CAR of 1.8% and a median CAR of 1.1%. These figures are slightly higher than those in Moeller, Schlingemann, and Stulz (2004), who find an average abnormal return of 1.4% for the combined firms between 1980 and 2001.

Figure 7 presents the distribution of the abnormal returns for the full sample during the event window of $[-1,1]$. Returns are positively skewed, and although the average deal has a positive abnormal return, about 40% of transactions exhibit negative returns over this horizon.

[Insert Figure 8 here]

Figure 8 plots the distribution of CARs over the shorter horizon, i.e., the $[-1,1]$ event window, for the acquirer and the target. Panel A shows the distribution of acquirers' CARs and Panel B plots the distribution of CARs for the target firms, both considering the full sample. Both distributions exhibit fat tails. Acquirer CARs exhibit low skewness, but target returns are positively skewed. When we restrict the sample to firms of similar size, measured by their enterprise value, acquirer returns show higher dispersion (Panel C), in contrast to target returns, whose distribution becomes more concentrated around positive abnormal returns (Panel D).

[Insert Table 2 here]

Table 2 presents additional sample cuts of the acquirer, target, and combined CARs. These cuts are intended to serve as a proxy for identifying firms that are very similar in terms of risk and cash-flow generation, as assumed in our model. Panel A only considers firms with similar enterprise value (the ratio of the larger firm to the smaller firm is less than or equal to 2). In Panel B, we restrict the sample to companies with comparable total book assets (the ratio of the larger firm to the smaller firm is at most 2). Panel C uses the same size definition as Panel B and further requires the firms to be in the same industry, according to the Fama-French 12 industry classification. In Panel D, we consider a subsample composed

of “mergers of equals”, defined as deals where: i) the target and acquirer agreed to a stock swap transaction; ii) the companies have approximately the same market capitalization; iii) the ownership of the new entity will be owned roughly 50/50 by the target and acquirer shareholders; and iv) both companies also have close to equal representation on the board of the new company. The average acquirer return in the different subsamples ranges from -0.1% to -2.3%, but is consistent with the average computed for the full sample. When firms are more similar, target returns are lower, suggesting that the acquisition premium increases with the relative size of the acquirer.¹²

[Insert Figure 9 here]

Finally, Figure 9 plots the distribution of combined CARs using the sample splits defined before. Consistently across the different panels, combined abnormal returns are on average positive, positively skewed, and with fat tails, with a significant mass of negative-return deals.

3.3 Calibration

Some parameters are chosen in a relatively direct fashion, as explained below:

- $\pi = 1$. This is just a normalization without any loss of generality. This means that we can interpret the realized level of x as the increase or decrease in the baseline BU profit rate.
- $\rho = 0.03$. The rate is expressed in annual terms and we use 3% as a level that is common in several quantitative models. Also, this rate delivers PE ratios that are

¹²In our sample, 92% of acquirers are larger than the target firm. The average acquirer is 68 times larger than the target firm (the median acquirer is about 6 times larger than the target firm).

quantitatively reasonable (taking into account the fact that our firms have no growth opportunities beyond one deal).

- $\lambda_1 = 0.04$, which implies an average life of 25 years. While it is debatable what is the exact number we should be targeting here, 25 is not extreme in either direction. Nevertheless, we conduct sensitivity analyses of our main results to changing this parameter (available from the authors upon request). We also note that using a λ_1 that is much higher than 0.04 will imply PE ratios that are too low.
- $\lambda_0 = 1/3$. This implies that merger opportunities arrive once every 3 years. This number was calibrated based on Anjos and Fracassi (2018).

The remaining steps/ingredients in the calibration are as follows:

- We assume that synergy opportunities are centered around zero and follow a truncated normal distribution with standard deviation σ_x , truncated at $\pm 4\sigma_x$.
- The bargaining process is governed by a low share s^l drawn from $\mathcal{S}$, which follows a triangular distribution on $[\underline{s}, 0.5]$ with mode 0.5.
- We use the mean, standard deviation, skewness, and fraction of negative values from the announcement return distribution to pin down the remaining parameters, σ_x , η , θ , and $\underline{s}$.

3.4 Targeted moments

A well-known concern with using announcement returns to proxy for merger value creation is that CARs capture not only deal-specific synergies but also information revealed about the merging firms' standalone values (Ben-David, Bhattacharya, Huang, and Jacobsen, 2025).¹³

¹³Announcement returns may also reflect market-driven price discovery that contributes to determining the realized synergy, rather than passively measuring a fixed primitive. This feedback channel (e.g., Bond,

In particular, this non-deal-related information can account for a substantial fraction of the announcement-day price movement, making raw CARs a noisy measure of synergies. However, Ben-David, Bhattacharya, Huang, and Jacobsen (2025) also document that the relative importance of deal-related information varies systematically with the size asymmetry between the merging firms: for transactions where the acquirer and target are of comparable size (relative size close to one), deal-related information accounts for roughly half of the variation in dollar-weighted CARs. We take this finding as additional motivation for restricting attention to deals between similar-sized firms (enterprise value ratio between 0.5 and 2), where the signal-to-noise ratio of announcement returns as a synergy proxy is most favorable. Furthermore, to isolate the synergy-related component of announcement returns more directly, we regress combined CARs on observable deal and firm characteristics—including relative size, industry relatedness, and method of payment—and use the moments of the resulting fitted-value distribution to guide our calibration exercise.

Specifically, we estimate the predictable component of combined CARs using two complementary approaches. First, we run a standard OLS regression of combined CARs on a set of deal and firm characteristics: log deal value, relative size, indicators for stock and cash payment, and indicators for tender offers, hostile deals, and competing bidders, the offer premium (with a missing-premium indicator); acquirer and target controls for log assets, leverage, book-to-market, and Tobin’s Q; acquirer Fama-French 12 industry dummies; and announcement-year fixed effects. Second, we estimate a random forest model using the same set of covariates, which allows for arbitrary nonlinearities and interactions among the predictors. We then extract the fitted values from each model— $\widehat{\text{CAR}}_i^{\text{OLS}}$ and $\widehat{\text{CAR}}_i^{\text{RF}}$ —and describe their distribution. Because the fitted-value distribution mechanically has lower variance than the raw CARs, we interpret the moments of this distribution as reflecting the

Edmans, and Goldstein (2012)) is outside our model, in which all participants observe x at the time of the match. Our calibration treats the fitted-CAR moments as informative about the moments of x on the assumption that this feedback contribution is second-order relative to the underlying synergy realization.

synergy-related component of announcement returns, net of idiosyncratic noise.

[Insert Figure 10 here]

Comparing the OLS and random-forest fitted values provides a useful diagnostic for whether the predictable component of announcement returns is sensitive to functional-form assumptions. Figure 10 plots the distribution of fitted values for the subsample of similar-sized firms (Panels A and B), and for the full sample (Panels C and D).

[Insert Table 3 here]

Table 3 reports the mean, the standard deviation, fraction of negative announcement returns, and skewness of the fitted CARs for each specification, restricted to the subsample of similar-sized firms (Panel A), and for the full sample of mergers (Panel B). These moments serve as the calibration targets for our quantitative exercise.

[Insert Table 4 here]

The results of our calibration are presented in Panels A and B in Table 4, for the two alternative target choices. Although the OLS and random-forest fitted values deliver similar means, some higher moments differ. We therefore use the OLS fitted moments as our baseline calibration targets and report the random-forest moments as a comparison. The inferred θ varies substantially (with the calibration for the full sample employing a lower θ by about 20 percentage points), which will affect the economic implications of our exercise, as we show in the next section.

[Insert Figure 11 here]

Finally, Figure 11 compares the model-implied distribution of announcement returns with the empirical distribution of fitted CARs for each calibration. The model-implied density tracks the empirical density reasonably well across the full support, capturing both the concentration of mass near zero and the positive skewness visible in the data. In both panels, the model produces a slightly flatter peak than the data, but matches the tails and overall shape.

3.5 Economic implications

In this section, we estimate how costly empire building is, and evaluate the magnitude of value-maximization spillovers.

Since the economy is populated by small and large firms, of EB and VM type, we first propose aggregate measures to evaluate the effects of EB merger policies. Consider a measure of aggregate profits and market capitalization of the economy defined in Lemma 2.

Lemma 2 *Aggregate profits AP are given by*

$$AP = a \pi + m_{vv} \frac{\mathcal{I}_{vv}}{\tilde{p}_{vv}} + m_{bb} \frac{\mathcal{I}_{bb}}{\tilde{p}_{bb}} + 2 m_{bv} \frac{\mathcal{I}_{bv}}{\tilde{p}_{bv}} \quad (30)$$

where

$$\mathcal{I}_{bv} := \frac{\mathcal{I}_{bv,b} + \mathcal{I}_{bv,v}}{2}, \quad \tilde{p}_{bv} := \frac{\tilde{p}_{vv} + \tilde{p}_{bv}^{hV}}{2} \quad (31)$$

Aggregate market capitalization AMC is given by

$$AMC = a_b J_{0,b} + a_v J_{0,v} + \frac{1}{\rho + \lambda_1} \left[m_{vv} \frac{\mathcal{I}_{vv}}{\tilde{p}_{vv}} + m_{bb} \frac{\mathcal{I}_{bb}}{\tilde{p}_{bb}} + 2 m_{bv} \frac{\mathcal{I}_{bv}}{\tilde{p}_{bv}} \right] \quad (32)$$

In our model, as EBs become more pervasive in the economy, their effect can be decomposed into two parts. On the one hand, EBs employ inefficient merger policies, which we classify as the direct effect of empire building. However, because VMs interact with EBs, the inefficiency of EBs is dampened or amplified by two counter-weighting forces: by being more selective, VMs impose a positive spillover effect on EBs' valuations, since a merger deal involving a VM and an EB prevents the latter from engaging in potentially inefficient deals with other EBs (the selection channel); but also impose a potential negative spillover effect on EBs' valuations in the presence of bargaining power asymmetries (the exploitation channel).

We start by characterizing the effects of empire building in light of our model, isolating the direct and indirect channels described before. Our measures are defined and derived in Proposition 2.

Proposition 2 *Define empire-building total cost Δ_{total} as the difference between a counterfactual market capitalization in a full-VM economy and the equilibrium aggregate market capitalization:*

$$\Delta_{\text{total}} := AMC|_{\theta=1} - AMC. \quad (33)$$

We define the direct cost of empire building Δ_{direct} as the difference between a counterfactual market capitalization in a full-VM economy and a counterfactual market capitalization in an economy where EBs and VMs are segregated:

$$\Delta_{\text{direct}} := AMC|_{\theta=1} - [\theta \cdot AMC|_{\theta=1} + (1 - \theta) \cdot AMC|_{\theta=0}]. \quad (34)$$

The indirect cost, defined as the difference between the empire-building total cost and the

direct cost, can be further decomposed into two parts:

$$\Delta_{\text{indirect}} := \Delta_{\text{total}} - \Delta_{\text{direct}} = \Delta_{\text{selection}} + \Delta_{\text{exploitation}}, \quad (35)$$

where $\Delta_{\text{selection}}$ captures a screening channel: by being more selective, VMs absorb EBs into high-synergy deals and remove them from the pool of potential EB-EB inefficient matches. Formally, $\Delta_{\text{selection}}$ is defined as the difference between a counterfactual segregated economy where VMs impose the same acceptance thresholds as in a full-VM equilibrium, and a counterfactual economy where VMs interact with EBs but still impose the same acceptance thresholds as in a full-VM equilibrium:

$$\Delta_{\text{selection}} := \left[\theta \cdot AMC|_{\theta=1, x^*} + (1 - \theta) \cdot AMC|_{\theta=0, x^*} \right] - AMC|_{\theta, x^*}, \quad (36)$$

where x^* denotes the acceptance thresholds in a full-VM economy.

$\Delta_{\text{exploitation}}$ is defined as the residual term between the indirect effect and the pure selection channel:

$$\Delta_{\text{exploitation}} := \Delta_{\text{indirect}} - \Delta_{\text{selection}}. \quad (37)$$

Finally, the exploitation channel is active only under bargaining-power asymmetry: $\Delta_{\text{exploitation}} = 0$ whenever $\underline{\varepsilon} \rightarrow \frac{1}{2}^-$ or $\eta = 0$, and at $\theta \in \{0, 1\}$.

Using the two calibrations from the previous section, we are able to compute global measures of empire-building costs and the spillover effects associated with the presence of VMs.

[Insert Figure 12 here]

Panel A in Figure 12 shows that, as expected, the costs of empire building grow as EBs become more pervasive (low θ). However, for intermediate levels of θ , a dampening effect

emerges from the presence of VMs, and the aggregate cost of EBs is substantially lower than it would otherwise be. Specifically, we estimate that for the calibration in Panel A of Table 4, the interaction between EBs and VMs reduces these aggregate costs by 31%. The difference between the costs EBs would generate in a counterfactual economy where EBs and VMs are segregated and the estimated costs is explained by the two indirect channels described before: on the one hand, the presence of VMs imposes positive spillover effects over EBs, by preventing bad deals that would otherwise occur; on the other hand, the inefficient EB merger policies generate an incentive for VMs to engage in rent-seeking behavior. These effects are computed separately in Panel B in Figure 12 for calibration 1. The selection effect reduces the costs and is quantitatively significant: in fact, the estimated total cost of empire building has about the same magnitude as these selection effects. Qualitatively, similar results are found for the calibration in Panel B of Table 4, as shown in Panels C and D of Figure 12. However, given the low estimated θ , this calibration implies higher costs of empire building, and a lower importance of the aforementioned indirect effects.

4 Conclusion

We propose a search-and-matching model of mergers to study an economy where some firms engage in empire building, accepting deals that are value-destroying, while other firms make decisions according to an optimal policy (value maximizers). We show that value maximizers generate two opposing spillover effects on empire builders. The first is a *selection* channel: by matching with empire builders, value maximizers absorb them into high-synergy deals and prevent them from engaging in future inefficient mergers, reducing the scope for value destruction economy-wide. The second is a *bargaining* channel: when surplus division is unequal, value maximizers can extract a disproportionate share of merger gains from empire builders, which lowers EB valuations. We take the model to data by calibrating four moments

of the equilibrium announcement-returns distribution to their empirical counterparts. In our calibration, selection spillovers are economically significant and positive, and more than offset the bargaining channel. As such, the negative impact of having EBs who apply inefficient merger policies is dampened.

More broadly, the economic mechanism we identify extends beyond the specific context of mergers and empire building. In any search-and-matching environment where some agents follow optimal decision rules and others rely on heuristics, the presence of optimizing agents can generate positive spillovers by absorbing heuristic types into efficient matches and thereby reducing the scope for future inefficient ones. This has implications for public policy: investments in financial literacy and general education that increase the share of value-maximizing agents in the economy may yield returns that are larger than a naïve calculation would suggest, precisely because of the indirect effects we quantify. For instance, in labor markets, firms that hire only when match quality is sufficiently high may generate similar positive spillovers by absorbing workers who might otherwise end up in poorly matched positions; analogous mechanisms could operate in marriage markets, where more deliberate individuals indirectly improve outcomes for less-selective participants, or in supply chains, where selective manufacturers prevent suppliers from entering inefficient long-term contracts with less-discerning buyers. As in our merger setting, however, these gains may be partly or fully offset wherever a bargaining channel lets the optimizing side extract surplus from the heuristic one; understanding the interaction between the direct gains of value maximizers and these indirect effects is thus crucial for evaluating the impact of public policy.

We contribute to the literature by studying a novel type of heterogeneity within a search-and-matching framework, and we believe the insights may carry over to other matching settings.

References

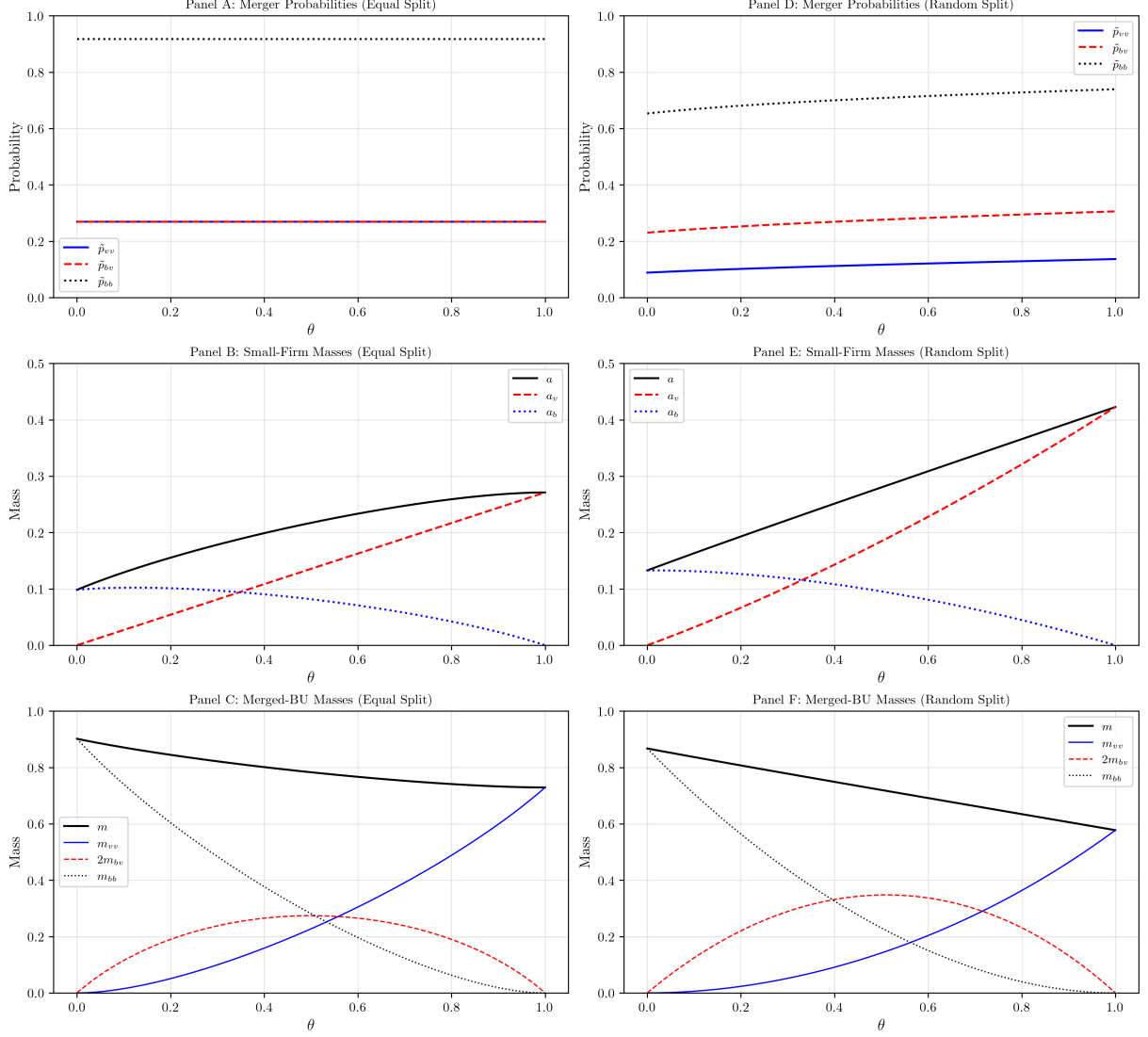
- Albuquerque, Rui, and Enrique Schroth, 2015, The value of control and the costs of illiquidity, *The Journal of Finance* 70, 1405–1455.
- Anjos, Fernando, and Cesare Fracassi, 2018, Technological specialization and the decline of diversified firms, *Journal of Financial and Quantitative Analysis* 53, 1581–1614.
- Barry, John W, Bruce I Carlin, Alan D Crane, and John R Graham, 2026, Hurdle rate buffers and bargaining power in asset acquisition, *Journal of Financial Economics* 178, 104240.
- Ben-David, Itzhak, Utpal Bhattacharya, Ruidi Huang, and Stacey Jacobsen, 2025, The (missing) relation between acquisition announcement returns and value creation (*forthcoming Journal of Finance*).
- Ben-David, Itzhak, and Alex Chinco, 2025, Capital budgeting for EPS maximizers (*NBER working paper 34061*).
- Bond, Philip, Alex Edmans, and Itay Goldstein, 2012, The real effects of financial markets, *Annu. Rev. Financ. Econ.* 4, 339–360.
- Custódio, Cláudia, Diogo Mendes, and Daniel Metzger, 2025, The impact of the financial education of executives on the financial practices of medium and large enterprises, *The Journal of Finance* 80, 2875–2920.
- David, Joel M., 2021, The aggregate implications of mergers and acquisitions, *The Review of Economic Studies* 88, 1796–1830.
- Dessaint, Olivier, B Espen Eckbo, and Andrey Golubov, 2025, Bidder-specific synergies and the evolution of acquirer returns, *Management Science* 71, 1391–1417.

- Diamond, Peter A., 1993, Search, sticky prices, and inflation, *Review of Economic Studies* 60, 53–68.
- Dong, Ming, David Hirshleifer, Scott Richardson, and Siew Hong Teoh, 2006, Does investor misvaluation drive the takeover market?, *The Journal of Finance* 61, 725–762.
- Gervais, Simon, James B Heaton, and Terrance Odean, 2011, Overconfidence, compensation contracts, and capital budgeting, *The Journal of Finance* 66, 1735–1777.
- Guadalupe, Maria, Veronica Rappoport, Bernard Salanié, and Catherine Thomas, 2024, The perfect match: assortative matching in mergers and acquisitions (*working paper*).
- Hackbarth, Dirk, and Erwan Morellec, 2008, Stock returns in mergers and acquisitions, *The Journal of Finance* 63, 1213–1252.
- Harford, Jarrad, 1999, Corporate cash reserves and acquisitions, *The Journal of Finance* 54, 1969–1997.
- Jensen, Michael C., 1986, Agency costs of free cash flow, corporate finance, and takeovers, *The American Economic Review* 76, 323–329.
- Jensen, Michael C., and William H. Meckling, 1976, Theory of the firm: managerial behavior, agency costs and ownership structure, *Journal of Financial Economics* 3, 305–360.
- Kőszegi, Botond, 2014, Behavioral contract theory, *Journal of Economic Literature* 52, 1075–1118.
- Malmendier, Ulrike, and Geoffrey Tate, 2008, Who makes acquisitions? CEO overconfidence and the market’s reaction, *Journal of Financial Economics* 89, 20–43.
- Moeller, Sara B, Frederik P Schlingemann, and René M Stulz, 2004, Firm size and the gains from acquisitions, *Journal of financial economics* 73, 201–228.

- Moeller, Sara B., Frederik P. Schlingemann, and René M. Stulz, 2005, Wealth destruction on a massive scale? A study of acquiring-firm returns in the recent merger wave, *The Journal of Finance* 60, 757–782.
- Mortensen, Dale T., and Christopher A. Pissarides, 1994, Job creation and job destruction in the theory of unemployment, *Review of Economic Studies* 61, 397–415.
- Rhodes-Kropf, Matthew, and David T. Robinson, 2008, The market for mergers and the boundaries of the firm, *The Journal of Finance* 63, 1169–1211.
- Shleifer, Andrei, and Robert W. Vishny, 2003, Stock market driven acquisitions, *Journal of Financial Economics* 70, 295–311.
- Smith, Lones, 2006, The marriage model with search frictions, *Journal of political Economy* 114, 1124–1144.
- Yim, Soojin, 2013, The acquisitiveness of youth: CEO age and acquisition behavior, *Journal of Financial Economics* 108, 250–273.

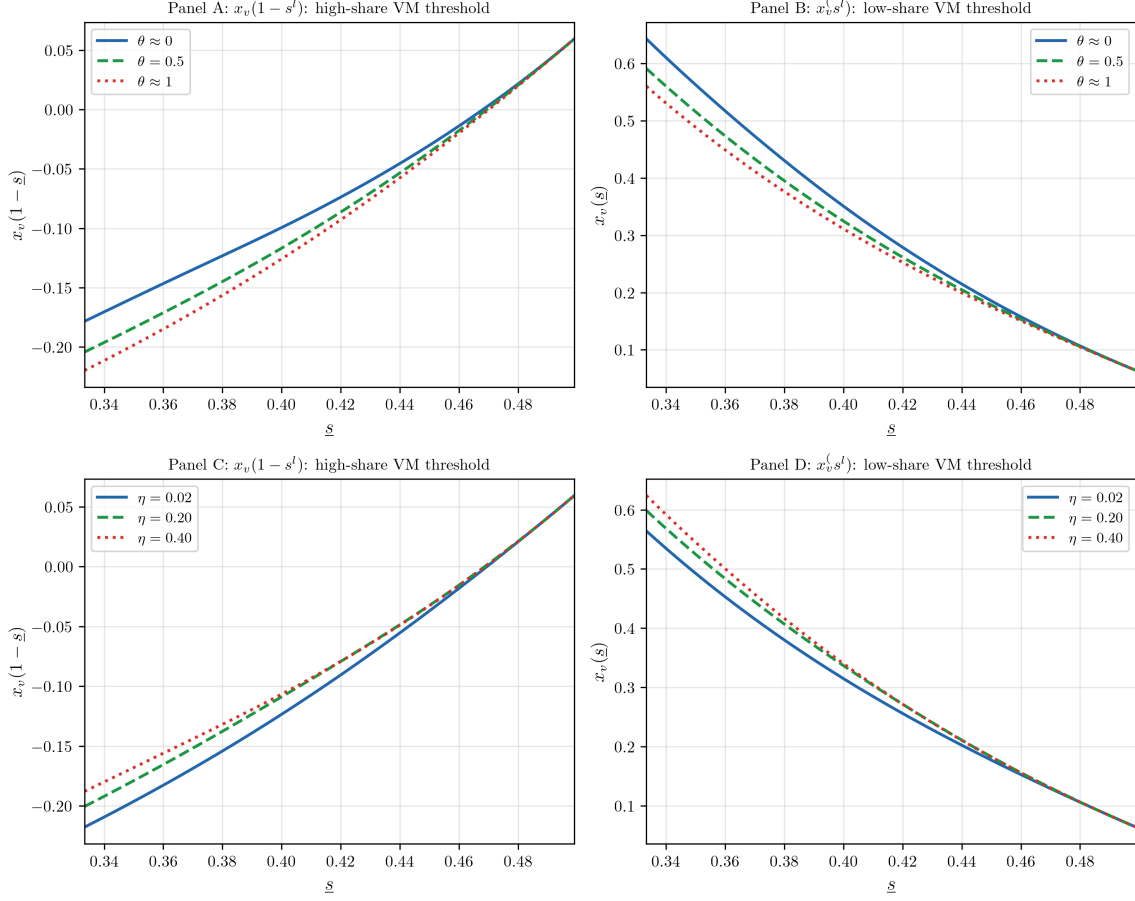
Appendix A: Figures and Tables

Figure 1: Merger Probabilities and Firm Composition for varying pervasiveness of value-maximizing types (θ)



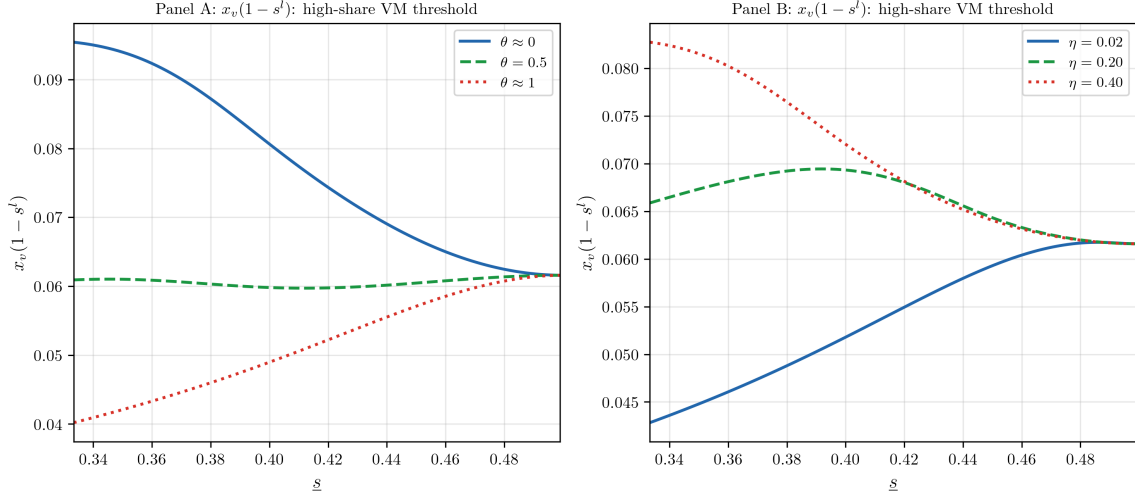
The figure plots merger probabilities, small-firm masses, and merged-BU masses for varying θ (fraction of VMs). Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(0, 0.10^2)$, on $[-0.40, 0.40]$. The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Panels A to C assume an equal split following a merger ($\underline{s} \rightarrow 0.5^-$); Panels D to F introduce random splits ($\underline{s} = 1/3$). Remaining parameters: $\pi = 1$, $\eta = 0.2$, $\rho = 0.05$, $\lambda_0 = 0.3$, and $\lambda_1 = 0.03$.

Figure 2: VM acceptance thresholds for varying share asymmetry ($\underline{s}$), evaluated at $s^l = \underline{s}$



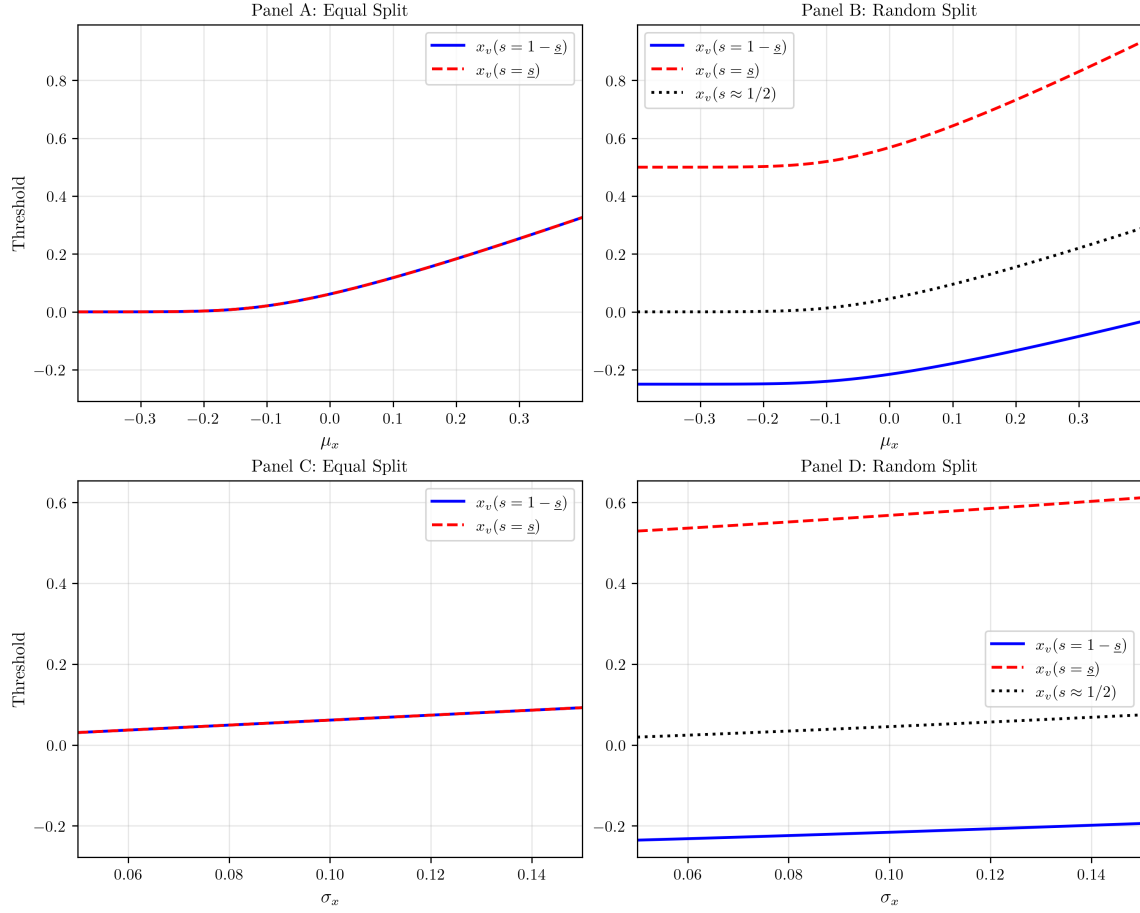
The figure plots the VM acceptance thresholds $x_v(s = 1 - \underline{s}; \underline{s})$ (when the VM receives the high share, Panels A and C) and $x_v(s = \underline{s}; \underline{s})$ (when the VM receives the low share, Panels B and D). Note that all policy functions are evaluated at the lower bound of the split distribution $\underline{s}$. Panels A and B vary the fraction of value-maximizing types $\theta \in \{0, 0.5, 1\}$, holding $\eta = 0.4$. Panels C and D vary the empire-building distortion $\eta \in \{0.02, 0.20, 0.40\}$, holding $\theta = 0.15$. Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(0, 0.10^2)$, on $[-0.40, 0.40]$. The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Remaining parameters: $\pi = 1$, $\rho = 0.05$, $\lambda_0 = 0.3$, and $\lambda_1 = 0.03$.

Figure 3: VM acceptance threshold for varying share inequality ($\underline{s}$), evaluated at $s^l \approx \frac{1}{2}$



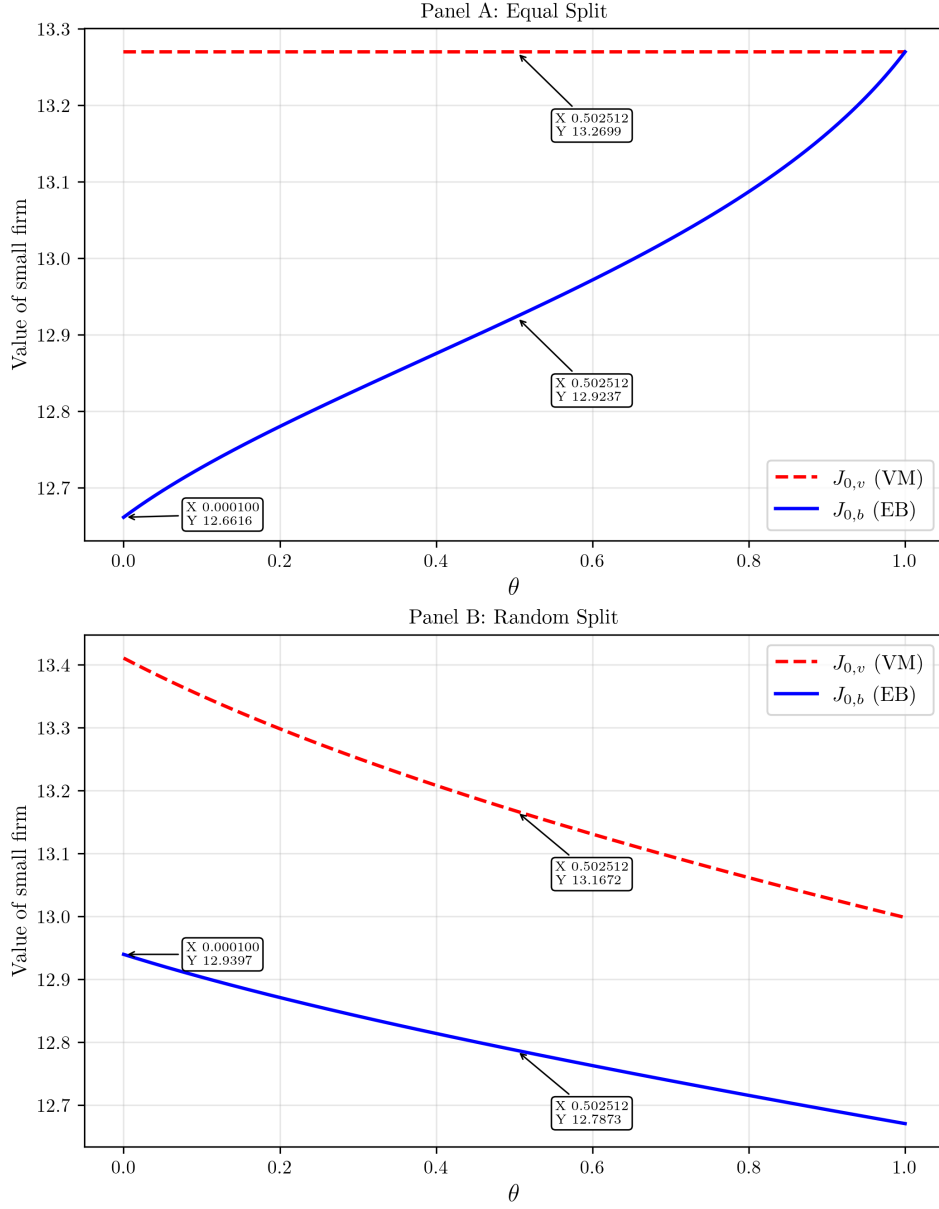
The figure plots the VM acceptance threshold $x_v(1-s^l; \underline{s})$ (evaluated at $s^l \approx 0.5$) as a function of the lower bound of the split distribution $\underline{s}$. Panel A varies the fraction of value-maximizing types $\theta \in \{0, 0.5, 1\}$, holding $\eta = 0.4$. Panel B varies the empire-building distortion $\eta \in \{0.02, 0.20, 0.40\}$, holding $\theta = 0.15$. Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(0, 0.10^2)$, on $[-0.40, 0.40]$. The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Remaining parameters: $\pi = 1$, $\rho = 0.05$, $\lambda_0 = 0.3$, and $\lambda_1 = 0.03$.

Figure 4: VM acceptance thresholds for varying synergy mean (μ_x) and dispersion (σ_x)



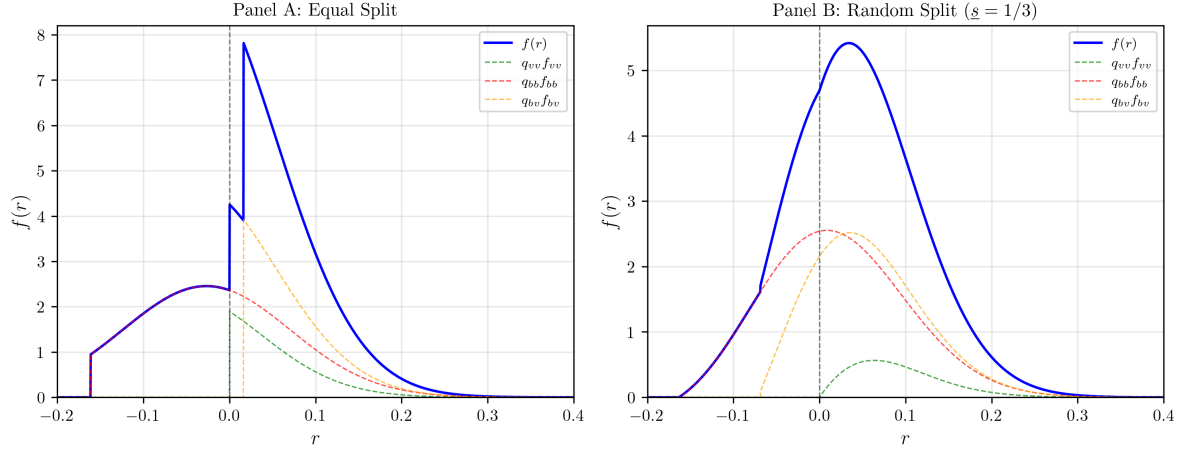
The figure shows the VM acceptance thresholds $x_v(1 - s^l)$ and $x_v(s^l)$, evaluated at different $\underline{s}$, for varying μ_x and σ_x . Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(\mu_x, \sigma_x^2)$, on $[\mu_x - 4\sigma_x, \mu_x + 4\sigma_x]$. In the top panels, $\sigma_x = 0.1$ and in the bottom panels $\mu_x = 0$. The left column assumes equal bargaining power ($\underline{s} \rightarrow 0.5^-$) and the right column assumes random splits ($\underline{s} = 1/3$). The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Remaining parameters: $\theta = 1/3$, $\pi = 1$, $\eta = 0.05$, $\rho = 0.05$, $\lambda_0 = 0.3$, and $\lambda_1 = 0.03$.

Figure 5: Value of small firms for varying pervasiveness of value-maximizing types (θ)



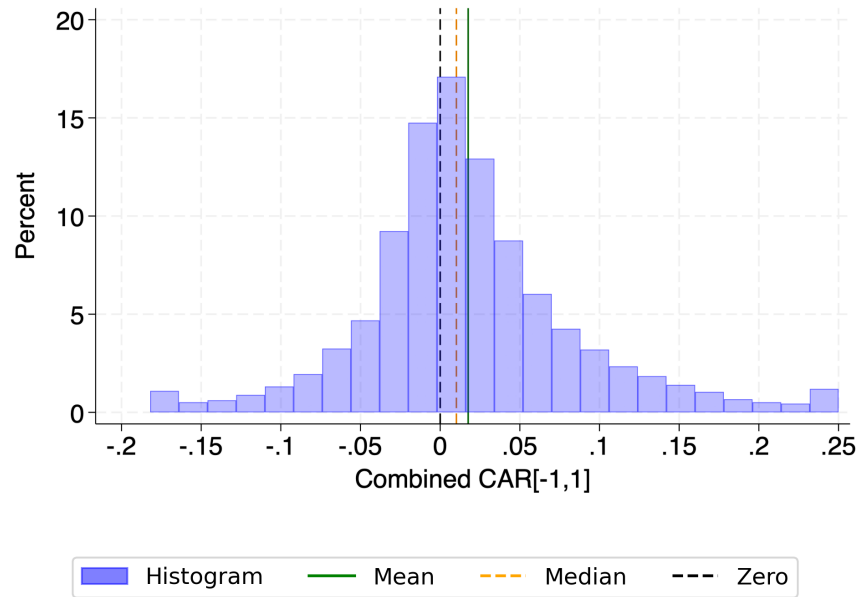
The figure plots the value of single-BU firms for varying θ (fraction of VMs), where the dashed red line refers to VMs ($J_{0,v}$) and the solid blue line to EBs ($J_{0,b}$). Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(0, 0.10^2)$, on $[-0.40, 0.40]$. The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Panel A assumes an equal split following a merger ($\underline{s} \rightarrow 0.5^-$); Panel B introduces random splits ($\underline{s} = 1/3$). Remaining parameters: $\pi = 1$, $\eta = 0.2$, $\rho = 0.05$, $\lambda_0 = 0.3$, and $\lambda_1 = 0.03$.

Figure 6: Distribution of announcement returns $f(r)$



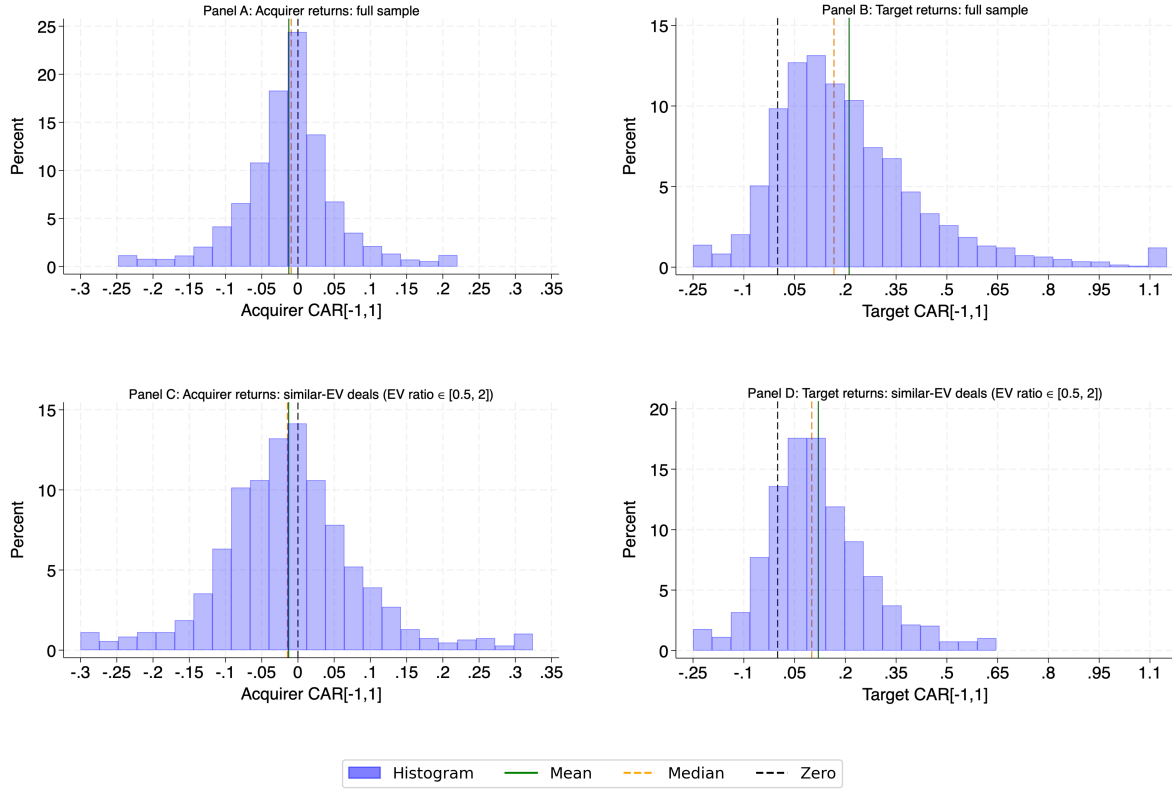
The figure shows the distribution of announcement returns characterized in Lemma 1. Synergies follow a truncated normal distribution, $\mathcal{X} = \mathcal{N}(0, 0.10^2)$, on $[-0.40, 0.40]$. The share distribution $\mathcal{S}$ is triangular on $[\underline{s}, 0.5]$ with mode 0.5. Panel A assumes equal bargaining power ($\underline{s} \rightarrow 0.5^-$); Panel B assumes random splits ($\underline{s} = 1/3$). Remaining parameters: $\eta = 0.2$, $\pi = 1$, $\rho = 0.05$, $\lambda_0 = 0.3$, $\lambda_1 = 0.03$, $\theta = 1/3$.

Figure 7: Distribution of combined cumulative abnormal returns (CARs) calculated over the event window $[-1, 1]$: Combined Firms



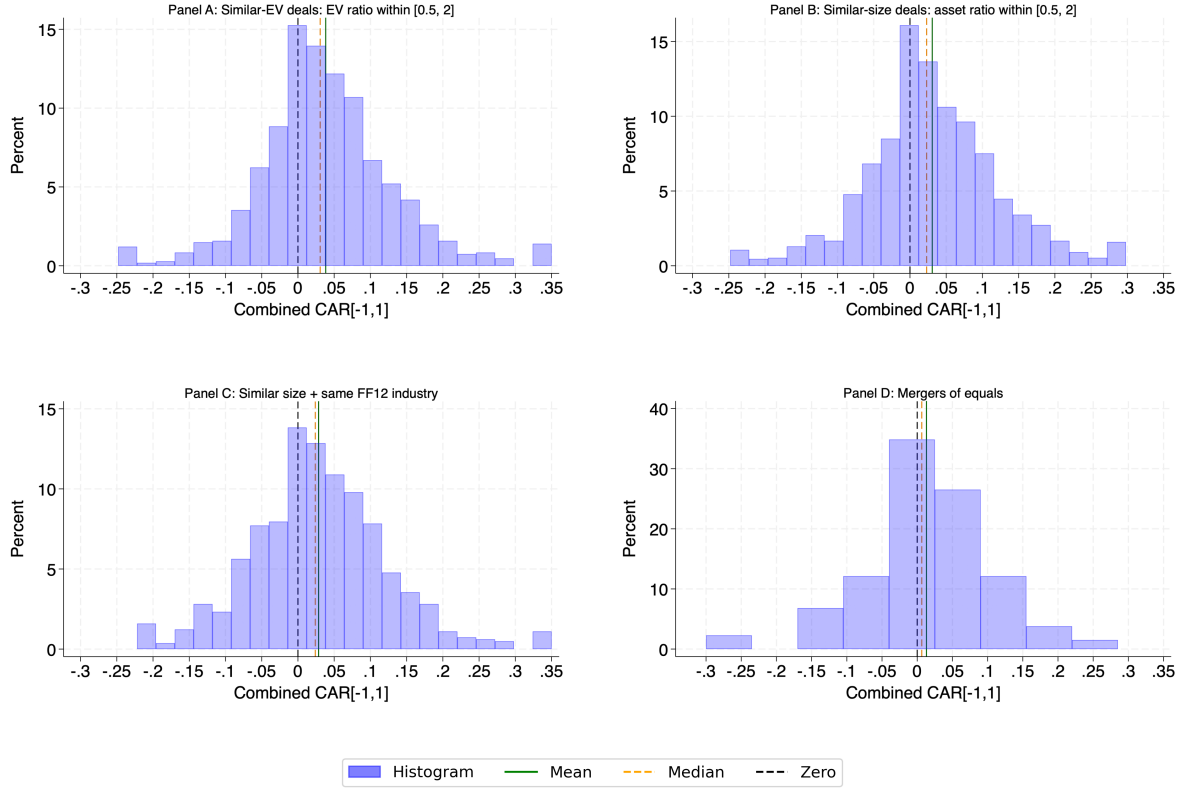
Distribution of the Combined Cumulative Abnormal Returns (CARs) calculated between one day before and one day after the announcement date. The combined CAR is computed as the value-weighted average of the acquirer's and target's CARs, using market capitalization as weights. The sample includes all deals with non-missing acquirer and target CARs. The vertical lines indicate the mean (solid green), median (dashed orange), and zero (dashed black). Returns are winsorized at the 1st and 99th percentiles.

Figure 8: Distribution of cumulative abnormal returns (CARs) calculated over the event window $[-1, 1]$: Acquirers and Targets



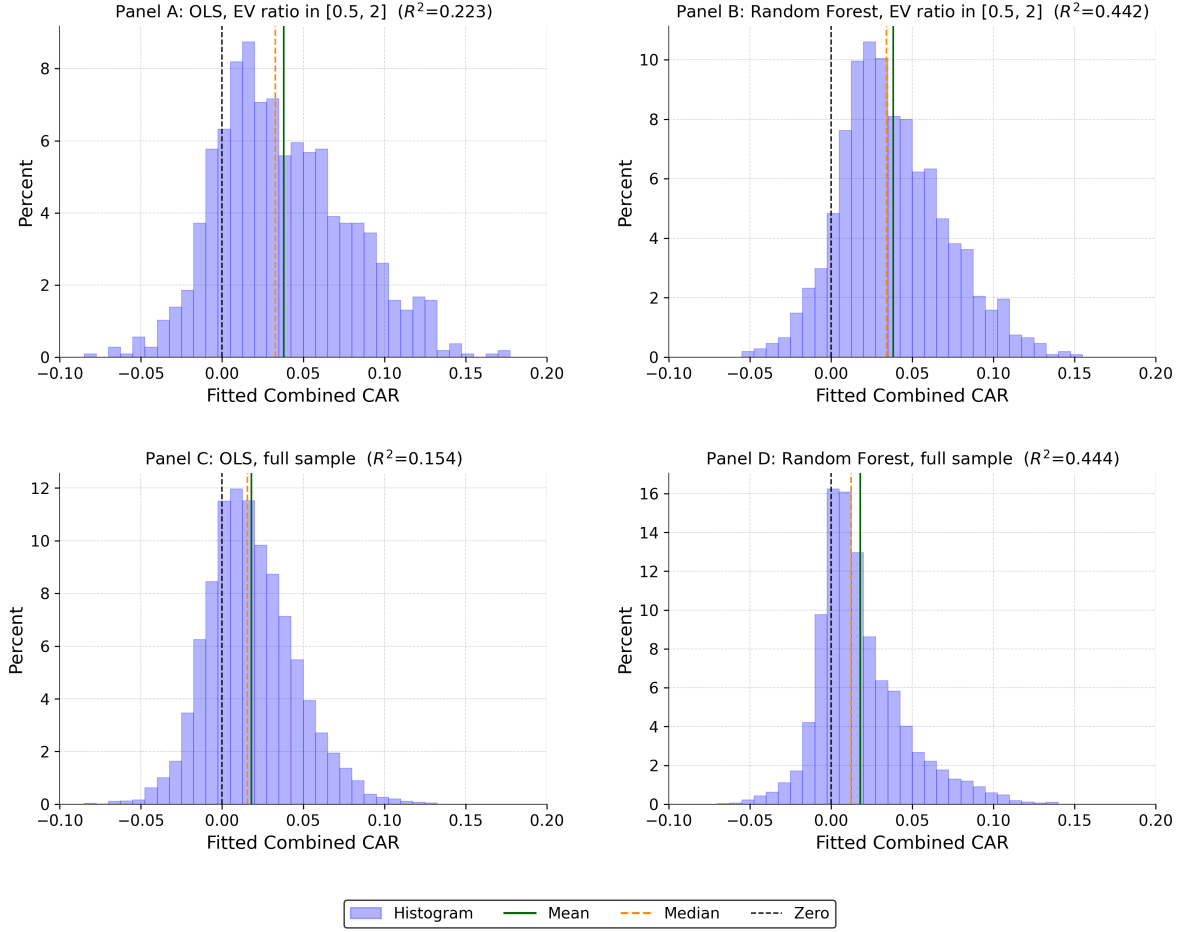
Distribution of the Cumulative Abnormal Returns (CARs) calculated between one day before the announcement date and one day after. Panel A depicts the distribution of the acquirer's CARs for the full sample, B shows the distribution of the target's CARs for the full sample, C shows the distribution of the acquirer's CARs restricted to similar-size deals where the enterprise value ratio between acquirer and target is within $[0.5, 2]$, and D shows the distribution of the target's CARs under the same similar-size restriction. The vertical lines indicate the mean (solid green), median (dashed orange), and zero (dashed black). Returns are winsorized at the 1st and 99th percentiles.

Figure 9: Distribution of combined cumulative abnormal returns (CARs) calculated over the event window $[-1, 1]$



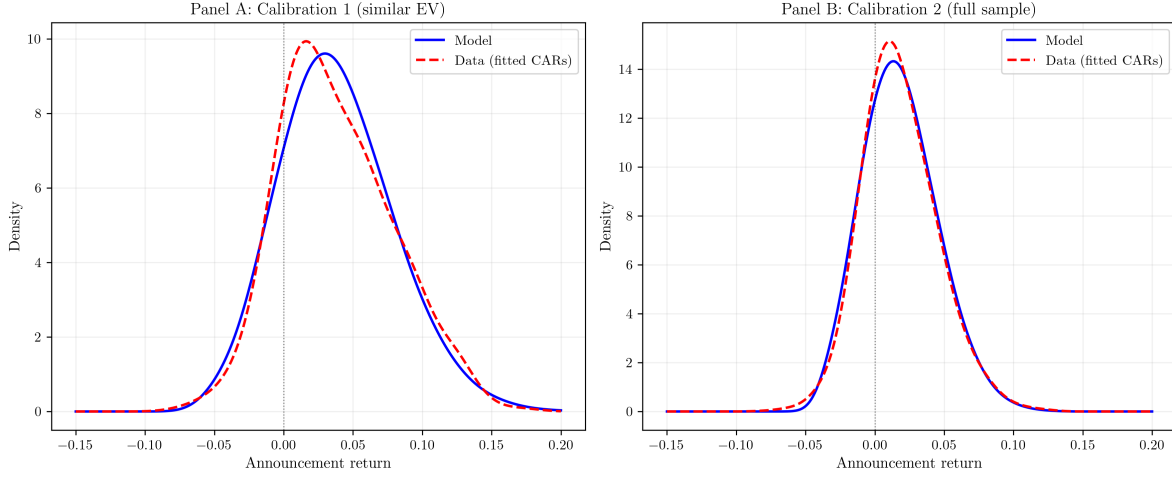
Distribution of the Combined Cumulative Abnormal Returns (CARs) calculated between one day before the announcement date and one day after, value-weighted by market capitalization of acquirer and target. Panel A shows the distribution for deals where the enterprise value ratio is within $[0.5, 2]$. Panel B depicts the distribution for deals where the asset ratio between acquirer and target is lower or equal to 2. Panel C includes deals where the asset ratio is at most 2 and both firms belong to the same Fama-French 12 industry. Panel D shows the distribution for mergers of equals, defined as deals where: i) the target and acquirer agreed to a stock swap transaction; ii) the companies have approximately the same market capitalization; iii) the ownership of the new entity will be owned roughly 50/50 by the target and acquirer shareholders; and iv) both companies also have close to equal representation on the board of the new company. The vertical lines indicate the mean (solid green), median (dashed orange), and zero (dashed black). Returns are winsorized at the 1st and 99th percentiles.

Figure 10: Distribution of fitted combined CARs: OLS vs Random Forest



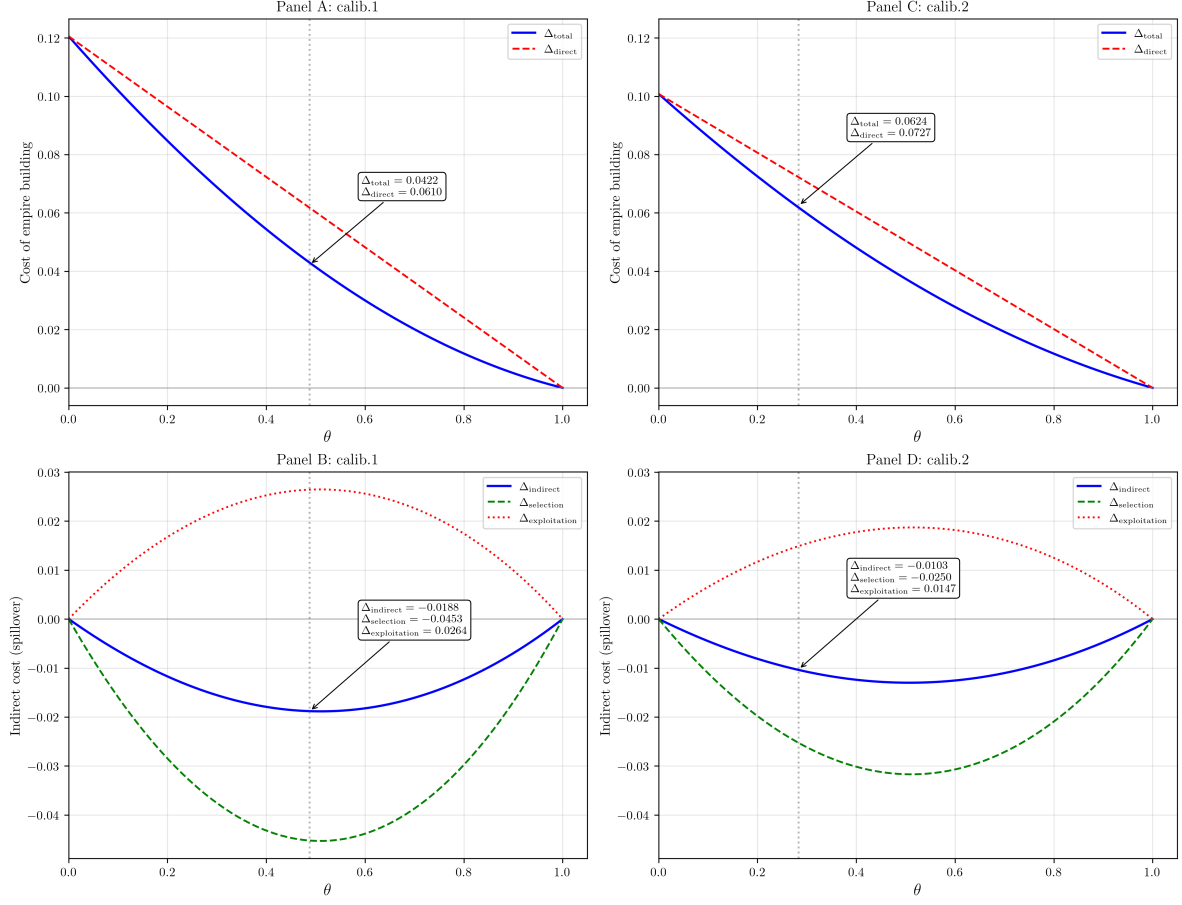
The figure shows the distribution of fitted values from regressing Combined Cumulative Abnormal Returns (CARs) on deal and firm characteristics using two methods: OLS (left column) and Random Forest (right column). The dependent variable is the combined cumulative abnormal return over the $[-1, 1]$ event window, computed as the value-weighted average of acquirer and target CARs. Covariates include deal controls (log deal value, relative size, payment method indicators, tender offer, hostile, competing bidder, and premium), acquirer and target firm controls (log assets, leverage, book-to-market, and Tobin's Q), acquirer Fama-French 12 industry dummies, and announcement year dummies. The Random Forest uses 500 trees with maximum depth of 20 and minimum leaf size of 5. Panels A and B restrict the sample to deals between firms of comparable enterprise value (EV ratio in $[0.5, 2]$); Panels C and D use the full sample. All fitted values are in-sample. The vertical lines indicate the mean (solid green), median (dashed orange), and zero (dashed black). The R^2 reported in each panel title corresponds to the in-sample R^2 of the respective model. All continuous variables are winsorized at the 1st and 99th percentiles.

Figure 11: Model-implied vs. empirical distribution of announcement returns



The panels compare the model-implied distribution of announcement returns (solid blue) with the empirical distribution of OLS fitted combined CARs (dashed red), using kernel density estimation. Panel A uses calibration 1, which targets the moments of fitted CARs for the subsample of similar-sized firms (enterprise value ratio in $[0.5, 2]$). Panel B uses calibration 2, which targets the fitted-CAR moments for the full sample of mergers. Please see Panels A and B in Table 4 for the parameter choice.

Figure 12: The costs of empire building



The panels decompose the costs of empire building as a function of θ . Panels A and C plot the total cost Δ_{total} (solid blue) and its direct component Δ_{direct} (dashed red), defined in equations (33) and (34). Panels B and D plot the indirect component Δ_{indirect} (solid blue) and its decomposition into the selection channel $\Delta_{\text{selection}}$ (dashed green) and the exploitation channel $\Delta_{\text{exploitation}}$ (dotted red), as defined in Proposition 2. The left column (Panels A and B) uses calibration 1, which targets the moments of OLS-fitted combined CARs for the subsample of similar-sized firms (enterprise-value ratio in $[0.5, 2]$). The right column (Panels C and D) uses calibration 2, which targets the fitted-CAR moments for the full sample of mergers. The vertical dotted line marks the calibrated value of θ , and the annotation box reports the magnitude of each measure at that point. Parameter values are reported in Panels A and B of Table 4.

Table 1: Summary Statistics of Cumulative Abnormal Returns

Panel A: CAR $[-1, 1]$									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	5,850	-0.013	0.069	-0.092	-0.045	-0.009	0.019	0.060	59.6%
Target	5,850	0.211	0.233	-0.020	0.055	0.166	0.317	0.503	12.9%
Combined	5,850	0.018	0.068	-0.054	-0.018	0.010	0.048	0.102	40.1%

Panel B: CAR $[-5, 5]$									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	5,846	-0.015	0.094	-0.126	-0.064	-0.013	0.033	0.088	57.6%
Target	5,846	0.235	0.250	-0.022	0.071	0.192	0.348	0.542	12.8%
Combined	5,846	0.018	0.092	-0.083	-0.032	0.012	0.065	0.128	42.8%

Panel C: CAR $[\text{Ann}-2, \text{Cl}+2]$									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	4,796	-0.044	0.259	-0.331	-0.161	-0.033	0.082	0.223	57.6%
Target	4,796	0.254	0.362	-0.137	0.037	0.214	0.429	0.688	20.8%
Combined	4,796	-0.006	0.237	-0.274	-0.120	-0.003	0.111	0.250	50.6%

The table reports summary statistics for Cumulative Abnormal Returns (CARs) computed using a market model with a value-weighted CRSP index and an estimation window of $[-361, -61]$ trading days relative to the announcement date. All CARs are winsorized at the 1st and 99th percentiles. *Combined* refers to the value-weighted average of acquirer and target CARs, weighted by their market capitalization in the month prior to the announcement. *% Neg.* is the fraction of observations with a negative CAR.

Table 2: Summary Statistics of Cumulative Abnormal Returns: Sample Splits

Panel A: Similar EV (EV ratio $\in [0.5, 2]$)									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	1,075	-0.013	0.096	-0.118	-0.069	-0.014	0.037	0.097	59.0%
Target	1,075	0.120	0.151	-0.042	0.024	0.101	0.206	0.318	19.2%
Combined	1,075	0.038	0.092	-0.060	-0.013	0.031	0.085	0.154	32.6%

Panel B: Similar size (asset ratio $\in [0.5, 2]$)									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	1,318	-0.019	0.093	-0.123	-0.073	-0.017	0.034	0.094	60.6%
Target	1,318	0.145	0.187	-0.042	0.026	0.117	0.235	0.367	18.7%
Combined	1,318	0.031	0.092	-0.076	-0.020	0.023	0.083	0.151	35.4%

Panel C: Similar size + same FF12 industry									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	817	-0.023	0.096	-0.137	-0.085	-0.020	0.038	0.098	60.6%
Target	817	0.154	0.195	-0.047	0.031	0.123	0.236	0.395	18.1%
Combined	817	0.028	0.094	-0.083	-0.028	0.024	0.083	0.145	37.0%

Panel D: Mergers of equals									
	N	Mean	SD	P10	P25	P50	P75	P90	% Neg.
Acquirer	132	-0.001	0.091	-0.097	-0.059	0.001	0.053	0.113	50.0%
Target	132	0.039	0.111	-0.073	-0.033	0.030	0.112	0.187	37.1%
Combined	132	0.013	0.092	-0.088	-0.025	0.006	0.065	0.119	43.9%

Distribution of the Cumulative Abnormal Returns (CARs) calculated between one day before the announcement date and one day after, restricted to subsamples of similar firms. Panel A restricts the sample to deals where the ratio of acquirer to target enterprise value (or its inverse) is below 2, where enterprise value is defined as market capitalization plus total debt (in book value) minus cash. Panel B restricts to deals where the ratio of acquirer to target book assets (or its inverse) is below 2. Panel C combines the size restriction from Panel B with the requirement that acquirer and target belong to the same Fama-French 12 industry. Panel D restricts to mergers of equals, defined as deals where: i) the target and acquirer agreed to a stock swap transaction; ii) the companies have approximately the same market capitalization; iii) the ownership of the new entity will be owned roughly 50/50 by the target and acquirer shareholders; and iv) both companies also have close to equal representation on the board of the new company. In all panels, CARs are winsorized at the 1st and 99th percentiles. *Combined* refers to the value-weighted average of acquirer and target CARs, weighted by market capitalization in the month prior to the announcement.

Table 3: Moments of the fitted combined CAR distribution: OLS vs Random Forest

Panel A: Similar EV (EV ratio $\in [0.5, 2]$)						
	N	Mean	SD	Median	% Neg.	Skewness
OLS	1,075	0.038	0.041	0.033	16.7%	0.393
Random Forest	1,075	0.038	0.033	0.034	9.3%	0.394

Panel B: Full Sample						
	N	Mean	SD	Median	% Neg.	Skewness
OLS	5,564	0.018	0.027	0.015	25.8%	0.391
Random Forest	5,564	0.018	0.027	0.012	23.0%	1.005

The table reports summary statistics of the fitted values from regressing Combined Cumulative Abnormal Returns (CARs) on deal and firm characteristics. The dependent variable is the combined cumulative abnormal return over the $[-1, 1]$ event window. Covariates include deal controls, acquirer and target firm controls, acquirer Fama-French 12 industry dummies, and announcement year dummies. The Random Forest uses 500 trees with maximum depth of 20 and minimum leaf size of 5. *Similar EV* restricts the sample to deals between firms of comparable enterprise value. *% Neg.* is the fraction of fitted values that are negative. All fitted values are in-sample. All continuous variables are winsorized at the 1st and 99th percentiles.

Table 4: Calibrated parameters

Panel A			
Description	Parameter	Value	Target
Discount rate	ρ	0.03	
Baseline BU profit rate	π	1.00	
Likelihood of merger matches	λ_0	1/3	
Likelihood of BU discontinuation	λ_1	0.04	
Synergy std. dev.	σ_x	0.0620	
EB merger threshold deviation	η	0.0755	
Fraction of VMs	θ	0.4880	
Lower bound of share split	$\underline{s}$	0.3950	
Mean ann. returns	$E[r]$	3.8%	3.8%
Std. dev. ann. returns	σ_r	4.1%	4.1%
Frac. negative returns	$\Pr(r < 0)$	16.7%	16.7%
Skewness ann. returns	$\text{Skew}[r]$	0.393	0.393
PE ratio	AMC/AP	14.47	

Panel B			
Description	Parameter	Value	Target
Discount rate	ρ	0.03	
Baseline BU profit rate	π	1.00	
Likelihood of merger matches	λ_0	1/3	
Likelihood of BU discontinuation	λ_1	0.04	
Synergy std. dev.	σ_x	0.0389	
EB merger threshold deviation	η	0.0530	
Fraction of VMs	θ	0.2832	
Lower bound of share split	$\underline{s}$	0.4354	
Mean ann. returns	$E[r]$	1.8%	1.8%
Std. dev. ann. returns	σ_r	2.7%	2.7%
Frac. negative returns	$\Pr(r < 0)$	25.9%	25.8%
Skewness ann. returns	$\text{Skew}[r]$	0.391	0.391
PE ratio	AMC/AP	14.38	

The table shows the parameters obtained from the steady-state model calibration. In each panel, four moments of the announcement-returns distribution are targeted: the mean, standard deviation, fraction of negative returns, and skewness. The targets correspond to the OLS-fitted combined CAR moments reported in Table 3. Panel A targets the fitted-CAR moments for the subsample of similar-sized firms (enterprise-value ratio in $[0.5, 2]$): mean of 3.8%, standard deviation of 4.1%, fraction of negative returns of 16.7%, and skewness of 0.393. Panel B targets the fitted-CAR moments for the full sample of mergers: mean of 1.8%, standard deviation of 2.7%, fraction of negative returns of 25.8%, and skewness of 0.391. The PE ratio is reported as a non-targeted moment.

Appendix B: Sample Construction

We employ data from the Thomson Reuters Securities Data Company (SDC) database to retrieve information from mergers and acquisitions (M&A) deals. We largely follow the restrictions applied in previous empirical work (e.g., Ben-David, Bhattacharya, Huang, and Jacobsen (2025)) for the sample construction. Hence, our dataset includes transactions where: (i) the announcement date is between January 1, 1980 and December 31, 2025, (ii) the acquirer is a U.S. company, (iii) the acquirer is a publicly traded firm, (iv) the deal is not classified as a spin-off, repurchase, self-tender, recapitalization, acquisition of partial or remaining interest, and the acquirer is not a leverage buyout firm, (v) the percentage of shares acquired (or sought, for non-completed deals) is at least 50%, (vi) the percentage of shares held by the acquirer six months before the announcement is less than 50%. Then, we (vii) drop duplicate deals in terms of the announcement and effective date, acquirer’s and target’s ultimate parent names, deal value, target and acquirer industry classification at the “mid-level”, target SIC code and % of stock as method of payment; and keep deals where (viii) the Center for Research in Security Prices (CRSP) database has stock data for the days around the announcement of the deal, Compustat has accounting data on the acquirer and target, and (ix) the deal value is non-missing in the SDC database.

Table 5: Sample Construction

Step	Filter Description	# of Deals		
		Completed	Not Completed	Total
1	Acquirer country: U.S.	375,070	78,600	453,670
2	Date announced: 1/1/1980 to 12/31/2025	374,389	78,356	452,745
3	Acquirer is public	130,886	55,566	186,452
4	Deal is not a spin-off, repurchase, self-tender, recapitalization, acquisition of partial or remaining interest, and acquirer is not an LBO firm	112,948	26,130	139,078
5	Percentage of shares acquired (or sought) is at least 50%	110,137	24,676	134,813
6	Percentage of shares held by acquirer six months before announcement is less than 50%	110,028	24,646	134,674
7	Drop duplicate deals in terms of announcement and effective date, acquirer and target ultimate parent names, deal value, target and acquirer industry classification, target SIC code and % of stock as method of payment	109,727	24,568	134,295
8	Require match to CRSP of both target and acquirer	5,564	1,701	7,265
9	Require match to Compustat of both target and acquirer	5,241	1,605	6,846
10	Require CAR[-1,1] measure to be non-missing for acquirer and target	4,857	1,475	6,332
11	Deal value is non-missing	4,807	1,043	5,850

This table reports the filters applied to the Thomson Reuters Securities Data Company (SDC) Domestic Merger and Acquisition database.

After applying these filters, we end up with a sample of 5,850 deals (including both completed and non-completed), as seen in Table 5. Importantly, we are also interested in the target's cumulative abnormal returns as we compute the combined CARs of the deal. Hence, our final sample is a combined one, where we consider deals that have the identifiers for both the target and the acquirer.

Appendix C: Proofs

Proof of Proposition 1. The low share s^l is drawn from $\mathcal{S}$ with support $[\underline{s}, 1/2]$. As $\underline{s} \rightarrow \frac{1}{2}^-$ the support shrinks to the single point $1/2$, so the limiting distribution is degenerate: $s^l \rightarrow 1/2$ and $s^h = 1 - s^l \rightarrow 1/2$ almost surely.

We start by rearranging equation (12), considering the value of $J_1(x)$ in equilibrium (equation (11)):

$$x_v(s) = \frac{J_{0,v}(\rho + \lambda_1)}{2s} - \pi. \quad (\text{C. 1})$$

Evaluating equation (C. 1) at $s^l = 1/2$,

$$x_v\left(s = \frac{1}{2}\right) = (\rho + \lambda_1)J_{0,v} - \pi =: \hat{x}(J_{0,v}), \quad (\text{C. 2})$$

so the high-share/low-share distinction disappears and there is a single VM threshold $\hat{x}$. For the EB, the shift condition (13) gives $x_b(s = \frac{1}{2}) = \hat{x} - \eta$, so the lower limit of the hybrid integral is $\max\{x_b(s = \frac{1}{2}), x_v(s = \frac{1}{2})\} = \max\{\hat{x} - \eta, \hat{x}\} = \hat{x}$, since $\eta \geq 0$. The binding party in a hybrid match is always the (more selective) VM, so the EB's slackness η never affects the threshold at which deals are accepted.

Reevaluating the merger-gain integrals from equations (19) and (21), every multiplier $s^l \times 2$

and $(1 - s^l) \times 2$ equals 1 and every lower limit equals $\hat{x}$, so

$$\mathcal{I}_{vv} \rightarrow \int_{\hat{x}}^{\bar{x}} (\pi + x) d\mathcal{X}(x), \quad \mathcal{I}_{bv,v} \rightarrow \frac{1}{2} \left[\int_{\hat{x}}^{\bar{x}} (\pi + x) d\mathcal{X} + \int_{\hat{x}}^{\bar{x}} (\pi + x) d\mathcal{X} \right] = \int_{\hat{x}}^{\bar{x}} (\pi + x) d\mathcal{X}. \quad (\text{C. 3})$$

The two hybrid sub-cases become identical, so $\mathcal{I}_{vv} = \mathcal{I}_{bv,v} =: \mathcal{I}(\hat{x})$. Likewise the acceptance probabilities coincide, $\tilde{p}_{vv} = \tilde{p}_{bv} = 1 - F_{\mathcal{X}}(\hat{x}) =: \bar{p}(\hat{x})$.

Substituting $\mathcal{I}_{vv} = \mathcal{I}_{bv,v} = \mathcal{I}(\hat{x})$ and $\tilde{p}_{vv} = \tilde{p}_{bv} = \bar{p}(\hat{x})$ into (17), both the numerator and denominator carry the common factor $a_v + a_b = a$, which cancels against the $1/a$ term. The composition of the matching pool drops out entirely, and the valuation reduces to

$$(\rho + \lambda_1) J_{0,v} = \pi + \lambda_0 \int_{\hat{x}(J_{0,v})}^{\bar{x}} (J_1(x) - J_{0,v}) d\mathcal{X}(x), \quad (\text{C. 4})$$

which does not depend on θ , η , or a_v, a_b . This is precisely the stand-alone HJB of a VM in an all-VM economy.

To see that (C. 4) pins down a unique value, define $G(J) := (\rho + \lambda_1)J - \pi - \lambda_0 \int_{\hat{x}(J)}^{\bar{x}} (J_1(x) - J) d\mathcal{X}(x)$, with $\hat{x}(J) = (\rho + \lambda_1)J - \pi$. The Leibniz term from differentiating the integral vanishes, since the integrand $J_1(x) - J$ is zero at the lower limit $x = \hat{x}(J)$ (by construction $J_1(\hat{x}) = (\pi + \hat{x})/(\rho + \lambda_1) = J$). Hence

$$G'(J) = (\rho + \lambda_1) + \lambda_0 [1 - F_{\mathcal{X}}(\hat{x}(J))] > 0, \quad (\text{C. 5})$$

so G is strictly increasing and (C. 4) admits a unique solution $J_{0,v}^*$. Because the equation involves neither θ nor η , this solution is identical for every $\theta \in [0, 1]$ and $\eta \geq 0$. This completes the proof. ■

Proof of Lemma 1. When a match of type (i, k) with synergy x is converted into a merger, the combined entity is worth $2J_1(x)$ irrespective of the realized split s^l , since

bargaining redistributes the merged value across the two parties but does not alter the cash flows it generates. The two units' pre-merger stand-alone value is $J_{0,i} + J_{0,k}$. Hence the combined announcement return is

$$r_{ik}(x) = \frac{2J_1(x) - J_{0,i} - J_{0,k}}{J_{0,i} + J_{0,k}}, \quad (\text{C. 6})$$

Because $J_1(x) = (\pi + x)/(\rho + \lambda_1)$ is continuous and strictly increasing in x , so is r_{ik} .

The total mass of BUs in big firms can be decomposed into:

$$m = m_v + m_b = (m_{vv} + m_{bv}) + (m_{bb} + m_{bv}) = m_{vv} + m_{bb} + 2m_{bv}. \quad (\text{C. 7})$$

Big firms have two BUs, so the fractions of big firms of each type are

$$q_{vv} = \frac{m_{vv}/2}{m/2} = \frac{m_{vv}}{m}, \quad q_{bb} = \frac{m_{bb}/2}{m/2} = \frac{m_{bb}}{m}, \quad q_{bv} = \frac{m_{bv}}{m/2} = \frac{2m_{bv}}{m}, \quad (\text{C. 8})$$

and they sum to one.

Consider a type (i, k) . Conditional on a type- (i, k) merger, the synergy x is drawn from $\mathcal{X}$ restricted to the set of realizations that are accepted—a set that depends on s^l and, in hybrid matches, on which party draws the low share—and the return is the image of x under the strictly increasing, continuous map r_{ik} . Consequently f_{ik} has as its support the interval $[r_{ik}(x_{ik}^{\min}), r_{ik}(\bar{x})]$, where $x_{ik}^{\min}$ is the smallest synergy that yields a type- (i, k) merger across all share draws (and splits). The upper endpoint is $r_{ik}(\bar{x})$ in every case, since the maximal synergy is accepted by all types in any equilibrium with positive merger flow. It remains to identify $x_{ik}^{\min}$.

vv matches. A *vv* deal requires the low-share VM to accept, i.e. $x \geq x_v(s^l)$. Since $x_v(\cdot)$ is decreasing, $\min_{s^l \in [\underline{s}, 1/2]} x_v(s^l) = x_v(1/2)$, so $x_{vv}^{\min} = x_v(1/2)$. The participation constraint

(12) at $s = 1/2$ gives $2 \cdot \frac{1}{2} J_1(x_v(1/2)) = J_{0,v}$, i.e. $J_1(x_v(1/2)) = J_{0,v}$, whence

$$r_{vv}(x_v(1/2)) = \frac{2J_1(x_v(1/2)) - 2J_{0,v}}{2J_{0,v}} = 0, \quad r_{vv}(\bar{x}) = \frac{J_1(\bar{x}) - J_{0,v}}{J_{0,v}}, \quad (\text{C. 9})$$

which is (26). In particular every vv deal carries a weakly positive combined return.

bb matches. A bb deal requires the low-share EB to accept, i.e. $x \geq x_b(s^l) = x_v(s^l) - \eta$.

Minimizing over s^l gives $x_{bb}^{\min} = x_b(1/2) = x_v(1/2) - \eta$, so

$$r_{bb}(x_v(1/2) - \eta) = \frac{J_1(x_v(1/2) - \eta) - J_{0,b}}{J_{0,b}}, \quad r_{bb}(\bar{x}) = \frac{J_1(\bar{x}) - J_{0,b}}{J_{0,b}}, \quad (\text{C. 10})$$

which is (27).

bv matches. A hybrid deal occurs in one of two equally likely sub-cases. If the VM draws the low share, the binding threshold is $x_v(s^l)$; if the VM draws the high share, both constraints may bind and the threshold is $\max\{x_b(s^l), x_v(1 - s^l)\}$. Since $x_b(s^l) = x_v(s^l) - \eta \leq x_v(s^l)$ and $x_v(1 - s^l) \leq x_v(s^l)$ (because $1 - s^l \geq s^l$ and x_v is decreasing), the high-share sub-case threshold never exceeds the low-share one; the smallest accepted synergy is therefore attained in the high-share sub-case,

$$x_{bv}^{\min} = \min_{s^l \in [\underline{s}, 1/2]} \max\{x_b(s^l), x_v(1 - s^l)\} =: x_{bv}^{\min}, \quad (\text{C. 11})$$

giving

$$r_{bv}(x_{bv}^{\min}) = \frac{2J_1(x_{bv}^{\min}) - J_{0,b} - J_{0,v}}{J_{0,b} + J_{0,v}}, \quad r_{bv}(\bar{x}) = \frac{2J_1(\bar{x}) - J_{0,b} - J_{0,v}}{J_{0,b} + J_{0,v}}, \quad (\text{C. 12})$$

which is (28). This completes the proof. ■

Proof of Lemma 2. Both aggregates sum over five BU cells: small EBs (a_b), small VMs (a_v), and merged BUs of type vv (m_{vv}), bb (m_{bb}), and bv ($2m_{bv}$). Small firms produce flow

π and are worth $J_{0,i}$, contributing $a\pi$ to AP and $a_b J_{0,b} + a_v J_{0,v}$ to AMC .

Since synergies are fixed at merger and big firms die at the common rate λ_1 , the steady-state distribution of x within each merged type is $\mathcal{X}$ truncated to that type's acceptance region. Hence the average per-BU profit $\pi + x$ for match type t equals $\mathcal{I}_t/\tilde{p}_t$, the ratio of the profit integral to the acceptance measure. For $t \in \{vv, bb\}$ this is immediate from (19)–(18) and (4). For $t = bv$, adding (20) and (21) the share multipliers in each region satisfy $s^l \cdot 2 + (1 - s^l) \cdot 2 = 2$, so the shares cancel and $\mathcal{I}_{bv} = \frac{1}{2}(\mathcal{I}_{bv,b} + \mathcal{I}_{bv,v})$ is the unweighted per-BU profit integral, paired with $\tilde{p}_{bv} = \frac{1}{2}(\tilde{p}_{vv} + \tilde{p}_{bv}^{hV})$.

Summing mass $\times$ average profit over cells gives (30). A merged BU is worth $J_1(x) = (\pi + x)/(\rho + \lambda_1)$, so its average value for type t is $\frac{1}{\rho + \lambda_1} \mathcal{I}_t/\tilde{p}_t$; adding stand-alone values gives (32). This concludes the proof. ■

Proof of Proposition 2. Equations (33)–(37) are definitions, and the two decomposition identities hold by construction. It remains to prove that $\Delta_{\text{exploitation}} = 0$ in the three stated cases. We first record that, substituting (33)–(34) into the definition of the indirect cost,

$$\begin{aligned} \Delta_{\text{indirect}} &= (AMC|_{\theta=1} - AMC) - \left(AMC|_{\theta=1} - [\theta AMC|_{\theta=1} + (1 - \theta) AMC|_{\theta=0}] \right) \\ &= [\theta AMC|_{\theta=1} + (1 - \theta) AMC|_{\theta=0}] - AMC. \end{aligned} \quad (\text{C. 13})$$

Equal splits $\left(\underline{s} \rightarrow \frac{1}{2}^-\right)$. By Proposition 1, the VM acceptance policy converges to the full-VM policy x^* for every θ and η ; the EB policy is then $x^* - \eta$ in every economy. Imposing x^* is therefore non-binding—the frozen-threshold economies coincide with the corresponding equilibria—so

$$AMC|_{\theta, x^*} = AMC, \quad AMC|_{\theta=1, x^*} = AMC|_{\theta=1}, \quad AMC|_{\theta=0, x^*} = AMC|_{\theta=0}. \quad (\text{C. 14})$$

Substituting these into (36) and comparing with (C. 13),

$$\Delta_{\text{selection}} = [\theta AMC|_{\theta=1} + (1 - \theta)AMC|_{\theta=0}] - AMC = \Delta_{\text{indirect}}, \quad (\text{C. 15})$$

hence $\Delta_{\text{exploitation}} = \Delta_{\text{indirect}} - \Delta_{\text{selection}} = 0$.

No distortion ($\eta = 0$). If $\eta = 0$, the EB acceptance policy $x_b(s) = x_v(s) - \eta$ coincides with the VM policy, so the two types are behaviorally identical and the economy is invariant to its composition: $AMC|_{\theta} = AMC|_{\theta=1} = AMC|_{\theta=0}$ for every θ , and likewise at the frozen thresholds. Every term in (33)–(36) is then zero, and in particular $\Delta_{\text{exploitation}} = 0$.

Endpoints ($\theta \in \{0, 1\}$). At $\theta = 1$ no EBs are present: $\Delta_{\text{total}} = AMC|_{\theta=1} - AMC|_{\theta=1} = 0$ and $\Delta_{\text{direct}} = AMC|_{\theta=1} - AMC|_{\theta=1} = 0$, so $\Delta_{\text{indirect}} = 0$; moreover $\Delta_{\text{selection}} = AMC|_{\theta=1, x^*} - AMC|_{1, x^*} = 0$, whence $\Delta_{\text{exploitation}} = 0$. At $\theta = 0$ the interacting economy is itself the all-EB economy, so by (C. 13) $\Delta_{\text{indirect}} = AMC|_{\theta=0} - AMC|_{\theta=0} = 0$, and $\Delta_{\text{selection}} = AMC|_{\theta=0, x^*} - AMC|_{0, x^*} = 0$; therefore $\Delta_{\text{exploitation}} = 0$. This concludes the proof. ■